

Script theory revis(it)ed: joke similarity and joke representation model*

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Abstract

The article proposes a general theory of verbal humor, focusing on verbal jokes as its most representative subset. The theory is an extension and revision of Raskin's script-based semantic theory of humor and of Attardo's five-level joke representation model. After distinguishing the parameters of the various degrees of similarity among the joke examples, six knowledge resources informing the joke, namely script oppositions, logical mechanisms, situations, targets, narrative strategies, and language, are put forward. A hierarchical organization for the six knowledge resources is then discovered on the basis of the asymmetrical binary relations, of the proposed and modified content/tool dichotomy, and, especially, of the hypothesized perceptions of the relative degrees of similarity. It is also argued that the emerging joke representation model is neutral to the process of joke production. The proposed hierarchy enables the concepts of joke variants and invariants, introduced previously by Attardo, to be firmed up, generalized, and augmented into a full-fledged taxonomy indexed with regard to the shared knowledge resource values (for example, two jokes may be variants on, that is, sharing, the same script oppositions and logical mechanisms). The resulting general theory of verbal humor is discussed in the light of its relations with various academic disciplines and areas of research as well as with the script-based semantic theory of humor, special theories of humor, and incongruity-based theories.

Introduction

The goal of the article is to outline a general theory of verbal humor as represented by verbal jokes.¹ The proposed theory postulates a hierarchi-

cal model of joke representation consisting of six levels and an indexed taxonomy of joke variance and invariance. Each level in the hierarchical model corresponds to and is determined by a knowledge resource, and each type of variance is indexed by one or more knowledge resources. Each knowledge resource is discovered as a parameter of joke difference, and this is where the article actually starts.

First, seven jokes are analyzed in terms of their degrees of similarity. Then the parameters of their similarities and differences are discussed in detail. While the discussion is informed by the script-based semantic theory of humor (Raskin 1985) and by the five-level model of joke representation (Attardo 1989), it goes well beyond both of these background sources.

In the central section of the article, the proposed theory is set up by a slow, complex, and painful procedure. First, a critical analysis of the earlier hierarchical model reveals its serious flaws and oversights. This analysis leads to the postulation of six knowledge resources informing the joke, and each resource corresponds to one of the parameters of joke difference discussed earlier in the paper.

The necessity and advantages of having a hierarchical model are discussed at a metatheoretical level, and the proposed theory is shaped and formatted by this discussion, which draws critically upon linguistic theory as a possible role model and a precedent and nonspecifically on the general principles of the philosophy of science. A modest contribution to the latter is proposed in the shape of two general principles of hierarchical ordering for levels of representation.

The subsequent four subsections establish the proposed hierarchy on the basis of the following three criteria:

- logical binary relations of dependence among the knowledge resources;
- the content- and tool-related nature of knowledge resources; and, most importantly,
- the hypothesized perceptions of the degrees of similarity among the main joke examples. It is also demonstrated that the proposed model is not a model of joke production and that, therefore, production-related considerations do not and cannot inform the model.

In the last section, the proposed model is summarized as a serious, even if friendly, revision and extension of both the script-based semantic theory of humor and the five-level model of joke representation. The theory of indexed joke variance and invariance is demonstrated to follow from the general theory. The academic disciplines informing the various

components of the theory are reviewed, and, finally, the status of the proposed theory *vis-à-vis* special theories of humor and the incongruity-based theories is briefly discussed.

The somewhat complex structure of the article is represented by the map in Figure 1. It is not clear to the authors at this point whether the map makes it easier to access the article or the article makes it easier to access the map.

Joke similarity

Many jokes are similar. Paraphrases and variants of the same joke can be found in print. People often retell jokes to each other, changing various aspects of them in the process. Let us consider a well-worn joke (1) along with six other jokes (2)–(7), each of which differs from (1) in one particular aspect.

- (1) How many Poles does it take to screw in a light bulb? Five. One to hold the light bulb and four to turn the table he's standing on (Freedman and Hofman 1980).
- (2) The number of Polacks needed to screw in a light bulb? Five — one holds the bulb and four turn the table (see Clements 1969: 22).
- (3) It takes five Poles to screw in a light bulb: one to hold the light bulb and four to turn the table he's standing on.
- (4) How many Irishmen does it take to screw in a light bulb? Five. One to hold the light bulb and four to turn the table he's standing on (see Raskin 1985: 176).
- (5) How many Poles does it take to wash a car? Two. One to hold the sponge and one to move the car back and forth (see Clements 1969: 22).
- (6) How many Poles does it take to screw in a light bulb? Five. One to hold the light bulb and four to look for the right screwdriver.
- (7) How many Poles does it take to screw in a light bulb? Five. One to take his shoes off, get on the table, and screw in the light bulb, and four to wave the air deodorants to kill his foot odor.

Thus, (1) and (2) are two slightly different ways of telling the same joke. The difference is in the choice of some words and of the syntactic constructions. In (2), *the number of* replaces *how many* in (1), *Polacks* is used instead of *Poles*, *needed for* *does it take*, etc. The last two sentences of (1) are made into one sentence with a dash in (2).

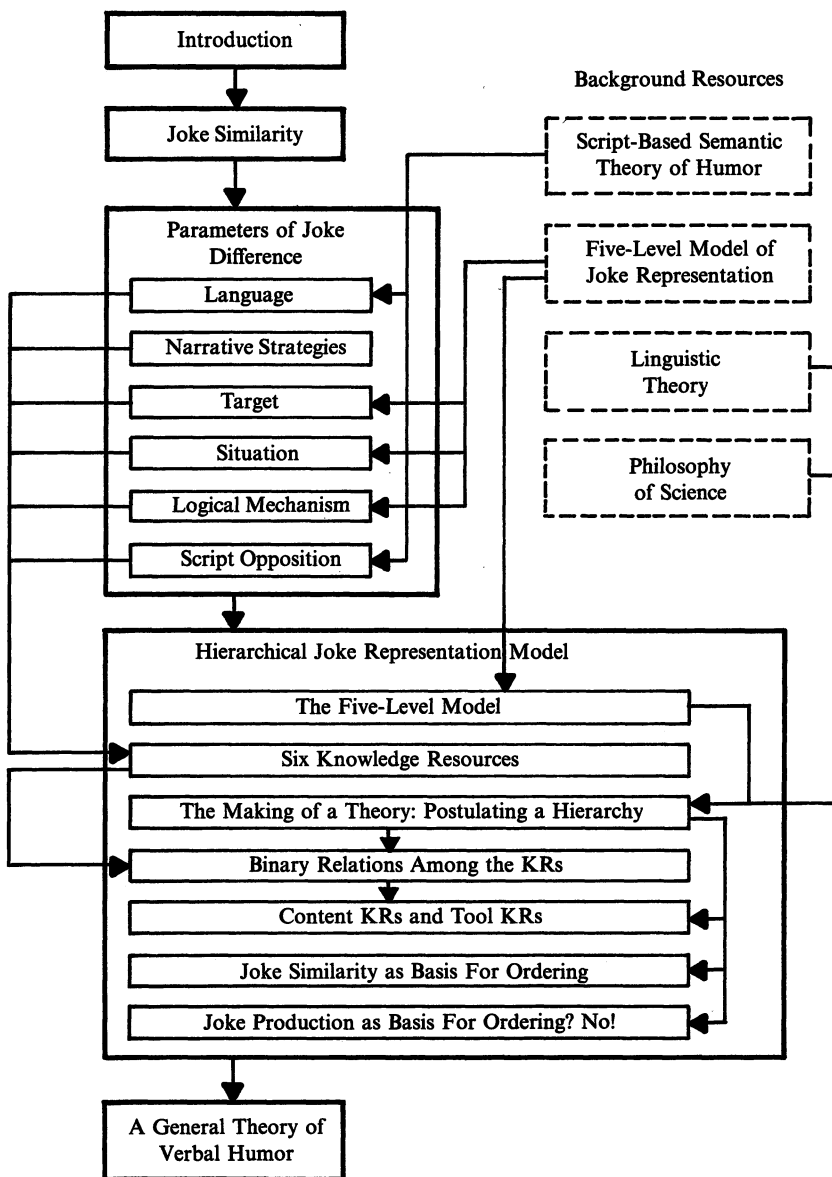


Figure 1. *Map of the article*

(3) uses exactly the same wording and much of the same syntax as (1), but it is not a riddle. Instead, it is a statement, an assertive expository text. (4) substitutes *Irishmen* for *Poles* and leaves the rest of (1) intact.

(5) is also about Poles doing something in an absurd fashion, but the action is different — it is no longer screwing in a light bulb but rather washing a car. (6) follows the wording of (1) up to the very end of the last sentence, where the other four Poles are suddenly made to look for a screwdriver instead of turning the table. In (7), one Pole screws in the light bulb in the normal fashion, but it takes four more to deodorize the air around his socks.

It seems that jokes (1) and (3) are very slight variations of each other. Joke (4) is directed against a different ethnic group but, otherwise, is identical. Jokes (5)–(7) introduce some changes. The situation in (5) shares with that of (1) the anomalous, stupid way of performing a simple chore and, moreover, the nature of the anomaly, namely, reversing the normal situation; normally, people turn the light bulb and move the sponge rather than turning the table and driving the car back and forth (or even funnier — heaving it to and fro).

In (6), this reversal is lost, and while the anomaly is there, it takes a different shape — looking for a screwdriver means treating a light bulb as a screw, which it is not. In (7), a totally different event takes place. There is absolutely nothing wrong in the joke with the way the Poles screw in a light bulb, but they are purported to be physically dirty.

It is clear that there is much more similarity among jokes (1)–(3) than between (1) and any one of jokes (5)–(7). Joke (4) is probably almost as similar to (1) as (2)–(4). Each of jokes (2)–(7) differs from (1) in a different way, along a different parameter. These parameters are discussed in more detail in the next section.

Parameters of joke difference

Parameter 1. Language

Joke (2) can be informally described as a mere paraphrase of (1). The difference in the choice of words, syntactic constructions, and other language options, including the division of the text into sentences, will be referred to as the difference in the language. Each joke can have hundreds and perhaps thousands of paraphrases because every sentence

in the text of the joke, just as any sentence of any natural language, may have multiple paraphrases. In fact, it has been calculated on the basis of various combinatorial possibilities that a sentence like *It was hard for Smith to translate the text because there were many technical terms in it* can be paraphrased in over 1,000,000 ways (see Mel'čuk 1974: 31).

The semantic competence of the native speaker of a language includes the ability to recognize paraphrases as such (see Katz and Fodor 1963: 486; Raskin 1985: 60). While some paraphrases may be more complex, verbose, or esoteric than others, it stands to reason that the hearer of the joke will consider all of them very similar. One good empirical criterion of that is that he or she will stop the teller in the middle, saying that the joke is familiar, or will not laugh at the end because, as is well known (see Fry 1963: 31–32), one may want to tell the same joke more than once but certainly not to hear it more than once.

In linguistic semantics, the notion of paraphrases is, of course, based on the equivalence of meaning.² In its extension to jokes, as used in the preceding two paragraphs, paraphrase often characterizes sequences of sentences rather than one sentence because a typical joke rarely consists of just one sentence (even the so-called “one-liners” usually have more than one sentence, albeit very brief — and quite often more than one line). The nature of the concept remains, however, the same: a joke paraphrase is the same joke.

The difference between jokes (1) and (2) and the evoked concept of paraphrase should clarify the nature of this parameter. It includes all the choices at the phonetic, phonologic, morphophonemic, morphologic, lexic, syntactic, semantic, and pragmatic levels of language structure that the speaker is still free to make, given that everything else in the joke is already given and cannot be tinkered with. What exactly is given will become clearer from the following five subsections of this section dealing with the other parameters of the joke, but basically, it is the content of the joke which has to be expressed within the parameter of language. We will see in the next section that the other parameters may limit or eliminate some of the choices from the parameter by insisting, for instance, on certain lexical choices (for instance, using a word like *Polish* in a Polish joke). It stands to reason to believe that more determination by the other parameters will be felt at the lexic, semantic, and pragmatic levels than at the levels less directly connected with meaning.

When an ordinary utterance is made, the content of what the speaker is going to say is roughly fixed in his or her mind, but the exact wording

has yet to be decided upon. In the process of making that decision, the speaker's entire language competence is activated at all levels, resulting in the activation of all the pertinent rules and the resulting placement of the appropriate units of various levels each in its place. Similarly, the parameter of language is responsible for the expression of the contents of the joke which include, along with the usual semantic material, a few specifically humorous elements and relations. An ordinary utterance belongs to casual language; a joke is noncasual. Noncasual language contains an additional layer of meaning, having to do with its specific function: to instruct in the case of a textbook, to entertain and to delight aesthetically in the case of fiction, to cause laughter in the case of humor. Thus, the parameter of language is responsible for the expression of casual meaning and, besides that, of a special joke-specific meaning determined by the other parameters.

One of the most important features of a joke is the punchline, so naturally, this parameter of language is responsible for the exact wording and placement of the punchline. According to Attardo et al. (forthcoming), most jokes place the punchline in the final position or a prefinal one, if followed by something inconsequential and anticlimactic. Raskin (1985: 114–117) distinguishes between two kinds of semantic script-switch triggers within the punchline, those based on an ambiguity and those based on a contradiction. Oring (1989) puts forward an important claim that it is the presence of the punchline which differentiates the joke from the funny story. Hetzron (1991) proposes a mixed-based classification of punchlines into “single-pulse,” or straightforward, and a large variety of “rhythmic,” or parallel-structured ones, for instance, joke (11) below.

Obviously, because of the crucial nature of the punchline, all the other parameters of the joke work toward it as well. It will be shown below, however, that the parameter of language finds itself in a unique relationship to all the others, namely, taking input from all of them. Because of that relationship, the placement of the punchline within the parameter of language, naturally without denying the contribution of the other parameters to the creation of the punchline, is fully justified. The arrangement of units and underlying rules of all levels of language structure is involved in the expression of the punchline, and as every joke teller knows, the punchline can go wrong at any level, including the “lowly” phonetics!³

Parameter 2. Narrative strategy

Joke (3) differs from (1) in the choice of a narrative strategy. By *narrative strategy*, for lack of a better term, we mean the genre, or rather microgenre as it were, of the joke, in other words, whether the text of the joke is set up as expository, as a riddle, as a question-and-answer sequence, and so on.⁴ Thus, (1) is a riddle or pseudoriddle, depending on whether one pauses between the first sentence and the rest of the joke waiting for the hearer's response or assumes that no response will be forthcoming. According to Esar (1952: 22–23), (1) is a conundrum, that is, a punning unguessable riddle; Pepicello and Green (1984) do not distinguish between conundra and genuine riddles.

(3) is a straightforward expository text. Various other strategies could be used to tell a joke. A sequence of questions and answers provided by the hearer used in (8) can be applied to (1) as well, resulting in (9). The triple sequence strategy, especially popular in the USA, is realized in (10); (11) results from its application to (1). A personal-ad format of (12) will transform (1) into (13).

- (8) Two Englishmen wander into each other in the middle of the Sahara desert after each has been lost for days. "British?" "British." "Oxford?" "Cambridge." "Queer?" "Queer." "Active?" "Passive." "Sorry, old chap, so long." "So long."
- (9) "Do you think one Pole can screw in a light bulb?" "No." "Two?" "No." "Three?" "No. Five. One to screw in a light bulb and four to turn the table he's standing on."
- (10) The triple tragic fate of a Chinese family: the father is a rickshaw, the mother is a geisha, and the son is Moishe.
- (11) How do the English, French, and Polish men screw in a light bulb? The Englishman looks at his watch and says, "Must dash to the pub in a minute. Just enough time to screw in that bloody light bulb." The Frenchman picks up the light bulb, shouts, "Un instant, chérie!" and hastily finishes the job. The Pole calls four friends to turn the table he will be standing on.
- (12) Middle-aged woman, plain, sick, poor. Wants to marry a Jew. [The Polish weekly *Przekrój*, which picked this ad up from a daily paper in 1956, wryly commented, "An antisemite."]
- (13) "Need help changing light bulb. Have bulb. Wanted: four strong men to turn table. Call Miaskowsky at 555-POLE."

An important aspect of this parameter has to do with the frequently used nonredundancy⁵ in the unveiling of the joke, resulting in missing links that the hearer must and can reconstruct. The wide use of the implicit in the text of the joke often borders on the deliberate violation of Grice's maxim of quantity (see Attardo 1990) and, as such, may be shared by this parameter with the preceding one. On the other hand, it may be also argued that the logical organization resulting from the use of the implicit and of the resulting inference activity bears on parameter 5 below. While this will not be studied further here, it will suffice to say that, for practical descriptive reasons,⁶ it would probably be preferable to treat this phenomenon within the narrative-strategy parameter of the joke.

Parameter 3. Target

Joke (1) deals with an absurdly stupid way of performing a simple and obvious task. As such, it can be targeted at any individual or group from whom such behavior is expected. These individuals or groups are referred to as the target of the joke. In the literature and personal experience, one runs into the same joke told of the Finns (Kerman 1980: 455), Newfoundlanders (Kerman 1980: 455), *carabinieri* (police) in Italy, Portuguese in Hawaii, West Virginians in Ohio, etc. As Davies (1990b) conclusively demonstrates, jokes like (1) travel widely around the world and are repeated in numerous similar situations.

The choice of the target is not completely free. It can only be someone for whom stupid behavior is believed to be natural and to require no explanation. In other words, the suitable target for the joke must have the "dumb" stereotype associated with it. It is interesting and important to note that the stereotype of dumbness should be associated with the targeted group totally independently of whether it corresponds to reality. In fact, it hardly ever does, being a sweeping generalization usually based on xenophobia, insecurity, ignorance, competition, etc. Moreover, it is unimportant if the teller or the hearer of the joke believes in the stereotype as long as they possess it and can apply it to the humorous act of telling and hearing the joke.

For most joke consumers, such a stereotype exists on the same fictional plane as unicorns, monsters, and Little Red Riding Hood (see Raskin 1985: 177–179 for further discussion). Whether the maintenance of such a stereotype casts any shadow on one's real-life attitude toward the

targeted group remains a controversial issue in the literature both on stereotypes and on humor (see Miller 1982; Samuels 1973; Zenner 1970; Spencer 1989; Schutz 1989; Davies 1990a). We firmly believe, however, that we have never been tempted to project the dumb-Pole stereotype, so significant in American humor, into reality.

It is also important to note right away that this parameter is optional for a joke, the only optional one among the six. Quite a few jokes do not have a clear target, for instance, the elephant jokes of the 1950s. Freud distinguished between the tendentious (targeted, in our terms) and nontendentious humor (1963 [1905]: 160–167); other scholars have tried to manipulate the terms “wit,” “humor,” “the comic,” etc., to capture the same distinction (see, for instance, Raskin 1985: 28 and references there).

Parameter 4. Situation

Joke (5) is also about an absurdly stupid way of performing a simple and obvious task. Like joke (1), it ascribes this behavior to the Poles, and it uses an identical sentence and syntactical structure. The activity is different, and the absurd way of doing it involves a different and appropriate set of actions. The principle underlying the two different activities in these jokes remains, however, the same: with the light bulb, the Pole will not turn it but rather hold on to it without moving his hand, and with the sponge, he does exactly the same thing. Obviously, many more activities can be substituted for these two ([14]–[17]), but in spite of all those differences in the numbers and specific actions, the resulting jokes preserve a considerable degree of similarity to (1):

- (14) How a Polack brushes his teeth? He holds the brush and moves his head (Clements 1969: 22).
- (15) How a Polack fans himself? He holds the fan and shakes his head (Clements 1969: 22).
- (16) How many Poles does it take to empty a car ashtray? Ten, to turn the car upside down (retargeted from a *carabinieri* joke recorded in Italy in the 1970s).
- (17) How many Poles does it take to drive a car? 500. One to drive and 499 to pull the road (Clements 1969: 22).

The choice of a suitable situation for a (1)-like joke is limited to simple

and obvious activities with universally known and widely practiced ways of going about them. An obscure and unknown activity will incapacitate the joke for most consumers:

- (18) How many Poles does it take to make an x-bar presentation of a sentence? 102 — one to think of a sentence, another to write it down, and 100 to look for a bar called “x” all over town.

More generally, every joke will be about something and thus will contain some “props.” These props constitute the situation of the joke. In most cases, as with joke (1), the activity constitutes the central element of the situation, which also includes, of course, the participants, objects, instruments, etc. It should be also noted that joke (18) is not as similar to joke (1) as are jokes (2)–(3) and a few others. The reason for that is that it is operated by a different logical mechanism.

Parameter 5. Logical mechanism

Joke (1) with all of its subsequent variations and deviations ([2]–[4], [9], [11], [13]) as well as jokes (5) and (14)–(17) are based on the same principle of the figure–ground reversal borrowed from gestalt psychology (see Bateson 1953, 1955;⁷ Talmy 1975, 1983). In the light-bulb situation, the ground is provided by the static environment, including, of course, the table or ladder used to reach the socket, and the figure is the bulb which should be screwed into the socket by being turned clockwise by the hand of the person doing it and standing on the table or ladder. Joke (1) reverses the roles by making the light bulb and the hand holding it static and making the environment rotate. The figure–ground reversal is the logical mechanism of joke (1) (see also Todorov 1978; Fonagy 1982 — compare Attardo 1988a; and, most recently, Hofstadter and Gabora 1989, whose “ur-joke” is a concept very similar to that of logical mechanism; see also Forabosco 1990b for a nice survey). It is indeed true that the method of changing the light bulb described in (1) will lead to the desirable result. The method is unusual and wasteful but successful, and it is fully justified logically. (We will discuss the unusual logic underlying joke [1] and other jokes at the end of this subsection.)

The same basic logical mechanism of figure–ground reversal in (5) is somewhat faultier logically. If the hand with the sponge remains completely static, the parts of the car below and above it as well as the front,

back, and roof will remain unwashed. The crudely approximate logic of a paralogism (see, again, Forabosco 1990b for a good survey) is at work there. The semblance of logic is accompanied by faulty or cheating inferential processes and will not withstand any close scrutiny. Jokes (16) and (19), the latter being the original, unadapted version of joke (2), are equally faulty: the contents of the ashtray, usually concealed below the dashboard, will actually remain in the car, thus defeating the purpose in (16), and, much more complicatedly, the light socket, sitting in an electric box attached to the joist above the ceiling, will not turn with it, and the static bulb will not be screwed in as in (19).

- (19) The number of Polacks needed to screw in a light bulb? Five — one holds the bulb and four turn the ceiling (Clements 1969: 22).

Joke (19) has another element, augmented in joke (17): it is rather unrealistic, though perhaps not totally impossible to turn the ceiling; it is, however, impossible to pull the road without breaking it into pieces. What we observe here is the addition of paralogical elements to the basic logical mechanism. If joke (1) is a pure and logical case of the instantiation of the mechanism, jokes (16) and (19) add to it elements of imprecision and sloppiness, which may actually avoid exposure without a careful analysis, and jokes (17) and (19) insert elements of the unreal which are much more obvious. The use of paralogisms instead of correct syllogisms in humor is quite permissible and compatible with the lack of commitment to the truth of the statement in the joke-telling mode of communication (see Raskin 1985: 55, 100–104) or, in other terms, with the willing suspension of disbelief, a requirement which, as already briefly mentioned in connection with unreal stereotypes, much humor shares with fiction.

The figure-ground reversal is only one of several logical mechanisms. Faulty logic, the paralogical elements, may occur by itself, without being associated with another mechanism. One of the best examples is perhaps one of Freud's favorites (usually, a dubious recommendation for a joke — see, however, Oring 1984) about the telepathic Rabbi of Cracow (1963 [1905]: 63):

- (20) In the temple at Cracow the Great Rabbi N. was sitting and praying with his disciples. Suddenly he uttered a cry, and, in reply to his disciples' anxious enquiries, exclaimed: "At this very moment the Great Rabbi L. has died in Lemberg." The community put on mourning for the dead man. In the course of the next few days people arriving from Lemberg were asked how the Rabbi had died

and what had been wrong with him; but they knew nothing about it, and had left him in the best of health. At last it was established with certainty that the Rabbi L. in Lemberg had not died at the moment at which the Rabbi N. had observed his death by telepathy, since he was still alive. A stranger took the opportunity of jeering at one of the Cracow Rabbi's disciples about this occurrence. "Your Rabbi made a great fool of himself that time, when he saw the Rabbi L. die in Lemberg. The man's alive to this day." "That makes no difference," replied the disciple. "Whatever you may say, the [telepathic vision] from Cracow to Lemberg was a magnificent one."

On the other hand, a simple reversal may not involve either the figure and ground or any paralogical elements. Called the "chiasmus" (see, for instance, Bloom 1965), it underlies the simple joke in (21) and many similar ones, a slightly more substantive one in (22), and a more complex joke ([23]), this one another of Freud's favorites:

- (21) Being honest isn't a question of saying everything you mean. It's a question of meaning everything you say (Milner 1972: 20).
- (22) Has anybody heard of a young black mugged in a Jewish neighborhood by four accountants?
- (23) Serenissimus was making a tour through his provinces and noticed a man in the crowd who bore a striking resemblance to his own exalted person. He beckoned to him and asked: "Was your mother at one time in service at the palace?" — "No, your Highness," was the reply, "but my father was" (Freud 1963 [1905]: 68–69).

Other logical mechanisms involve analogy ([24]), which may be complicated by paralogical elements to create somewhat false ([25]) or totally false analogy ([6], [26]):

- (24) George Bush has a short one. Gorbachev has a longer one. The Pope has it but does not use it, Madonna does not have it. What is it? A last name (see *HUMOR* 4: 1).
- (25) Have you heard of the Pole who ordered a swimming pool with a sliding driveway, so that he could wash his car in it and save on the cost of car washes?
- (26) A student is failing an oral exam in entomology. Finally, the professor shows him the leg of an insect and asks to identify the owner. The student is unable to do that, and the professor flunks

him. As the student is leaving the room the professor realizes that he did not put down the student's name. "What's your name, young man?" he shouts after the student. The student sticks his leg back into the room and says, "You guess it, professor."

Joke (24) exhibits an additional phenomenon known in linguistics as "false priming" or "garden path." As in the sentence *The astronomer married a star*, where the comprehension is hampered by the fact that astronomer suggests, or primes, an inappropriate meaning of star (the celestial-body meaning instead of the heavenly!), the hearer of joke (24) is led down the garden path toward the obscene antecedent of "it," that is "penis," even though he or she has no idea if the statements on Bush and Gorbachev are correct (no commitment to the truth of the humorous statements illustrated again). Joke (27) is a much simpler example of the garden path phenomenon.

- (27) Should a person stir his coffee with his right hand or his left hand? Neither. He should use a spoon (Esar 1952: 21).

Joke (27) implements the garden-path mechanism by manipulating the acceptable level of the obvious: the question seems to assume that the hand holds the spoon, and the hearer takes it for granted. But then the tricky answer denies the assumption; the hearer backtracks on the text of (27) and discovers that, indeed, it never said that the spoon was there, in the hand, either left or right. It is shown elsewhere (Raskin 1990a, 1990b) that similar manipulations of the obvious may lead to extremely sophisticated jokes.

The most trivial logical mechanism, a kind of default, is the juxtaposition of two different situations determined by the ambiguity or homonymy in a pun, such as Esar's feeble jokes ([28]–[29]) or a post-*glasnost* Soviet joke ([30]) ridiculing Gorbachev's geriatric predecessors:

- (28) Why does a donkey eat thistles? Because he's an ass (Esar 1952: 63).
 (29) The first thing which strikes a stranger in New York is a big car (Esar 1952: 77).
 (30) Who supports Gorbachev? Oh, nobody. He is still able to walk on his own.

Not every pun is a feeble joke (see Marino 1988). Combined with a nontrivial logical mechanism rather than merely depending on whatever meanings the main word accidentally brings together, a pun can be a

decent joke as well. And a mere juxtaposition can be brought about even without a pun as, for instance, in the "Gobi Desert Canoe Club" T-shirt.

It has been frequently noted in humor research, especially since the inception of the incongruity theories of humor (see Raskin 1985: 31–36 and references there), that a joke must provide a logical or pseudological justification of the absurdity or irreality it postulates. Very little has been done in humor research to follow up on this observation. One exception has been Aubouin (1948);⁸ another, apparently an independent one, has been Ziv (1984).

Ziv calls the faulty logic used in jokes *local logic*. "Local logic is appropriate only in certain places ... because it brings some kind of explanation to the incongruity" (Ziv 1984: 90). While his "partial suitability" of the local logic of a joke is parallel to Aubouin's "momentary, superficial" justification, Ziv notes also that the local logic explanation works only "if we are willing to play along" — an important element that Aubouin missed entirely.

Raskin's (1985) script-based semantic theory of humor did not address logical mechanisms directly. It did, however, brush over them in the discussion of both script oppositions and triggers. The general actual vs. nonactual type of script opposition usually implies that the former member of the pair is most obviously introduced by the text of the joke, at least initially. The nonactual script is usually not "really there" by any ordinary logic and, therefore, has to be pseudologically justified.

Parameter 6. Script opposition

Joke (7) definitely stands out among all the Polish jokes used so far because it is not about dumbness. Instead, it is about being dirty. Historically, American Polish humor did use this purported trait about Poles (see, for instance, Davies 1990b: 84–101), as most xenophobic humor does in conjunction with purported dumbness. It did, however, yield to purported dumbness long ago, and for a long time now in the USA, a Polish joke has been a joke about dumbness or a dumb joke (see Raskin 1985: 185–189 and references there; see also Davies 1990b: 100–101).

In terms of the script-based semantic theory of humor (SSTH), jokes (1)–(6) and many subsequent examples evoke the script of dumbness while joke (7) fails to do so, evoking the script of dirt instead. A chunk of structured semantic information (see Raskin 1986), the script can be

understood for the purposes of this article as an interpretation of the text of a joke. The main claim of SSTH is that the text of a joke is always fully or in part compatible with two distinct scripts and that the two scripts are opposed to each other in a special way. In other words, the text of the joke is deliberately ambiguous, at least up to the point, if not to the very end. The punchline triggers the switch from the one script to the other by making the hearer backtrack and realize that a different interpretation was possible from the very beginning.

SSTH postulated three levels of script opposition (without actually saying so explicitly). First, at the most abstract level, the joke opposes the *real* to the *unreal*, that is, factual reality to an imagined one. This may take three possible forms, existing at a lower level of abstraction, namely, the *actual* vs. *nonactual*, *normal* vs. *abnormal*, and *possible* vs. *impossible*. At the lowest level of abstraction, these three can be manifested by such oppositions as *good* vs. *bad*, *life* vs. *death*, *sex* vs. *nonsex*, *money* vs. *no-money*, *high stature* vs. *low stature*, etc.

While the list above was not meant to be exhaustive, it turns out to be quite representative because the number of oppositions exhibited by jokes is finite and limited. It has become clearer since the inception of SSTH that the oppositions of the lowest level are not equal in generality. Thus, life/death, no-sex/sex, money/no money, and high/low stature can all be seen as subcases of good/bad. Good/bad may be then alternatively treated as the fourth type of opposition of the higher level or as a subtype of normal/abnormal. If the latter is preferred, then good/bad will remain an opposition of the third level of abstraction, which will then become the second lowest rather than the lowest one, with quite a few subtypes of good/bad occupying a lower, fourth level.

Nondumb/dumb is also of the normal/abnormal and good/bad type of opposition, and so is clean/dirty. The two oppositions are still different, and jokes (1), with all of its versions, on the one hand, and (7), on the other, based on these two different oppositions, are less similar than those jokes which are all based on the same nondumb/dumb opposition.

Can it be concluded then that the shared script opposition assures a higher degree of similarity than the other shared parameters? If so, it would follow that the script parameter is the most basic of all. SSTH made this assumption, basically by ignoring the other five parameters, except that it can be claimed that its treatment of the script-switching triggers in the punchline came very close to the parameter of logical mechanisms. Even if this claim is granted, however, SSTH distinguished

between only two types of triggers, ambiguous and contradictory, and thus did not go very far into the study of logical mechanisms as such. SSTH would view them essentially as a mere implementation of the script opposition, the concept which was basic to the theory and with which it was fully preoccupied. It is true, though, that the discussion of a few particular cases of those two types of triggers did touch on some classes of jokes mentioned here in connection with parameter 5.

There is no easy answer to the question opening the previous paragraph. The hierarchy and level of generality of the parameters are very complex issues, and they are discussed in detail in the next section. The outcome of that discussion will shape the theory based on the parameters and constitute a major revision of SSTH.

Hierarchical joke representation model

The five-level model

Attardo (1987) postulated the existence of different levels of abstraction in the representation of the text of a joke. The idea was later (1988b, 1989) connected with the SSTH script opposition, which was presented as the most abstract form of the representation of a joke.

The later version posited five levels of joke representation. The first and lowest level was the actual text of the joke. The second level assigned a specific language form to the prelanguage representation of the joke. The third level assigned a target to the joke. At the fourth level, the template of the joke was created by combining the script opposition and logical mechanism of the joke. And the fifth, most abstract level contained both the script oppositions and logical mechanisms.

All of this becomes much clearer in the top-down version, illustrated with an example (Table 1). At the fifth level, the smart/dumb opposition is present among the others; similarly, the figure-ground reversal mechanism will be found there along with other logical mechanisms. At the fourth level a joke template is created by combining the smart/dumb opposition with the figure-ground reversal logical mechanism. At the third level, the Poles are selected as the joke target on the basis of complex cultural knowledge, including fictional stereotypes, and the situation of light-bulb changing is implicitly introduced. At the second level, the resulting representation is assigned a specific language expression, com-

Table 1. *Attardo's five-level joke representation model*

Level no.	Name	Contents, action, choice(s), or result
5	Basic	Script opposition (smart/dumb) and logical mechanism (figure-ground)
4	Template	Juxtaposing: smart/dumb and figure-ground reversal
3	Target + Sit.	Selected: target = Poles; situation = light-bulb changing
2	Language	Selected: words, syntax, sentence lineup, etc.
1	Surface	Result: text of joke (1)

plete with the syntax and choice of words. The first level, the result of this activity at the preceding level, is joke (1).

The ordering of the levels is intuitively clear for a linguist because it follows the meaning-to-sound scheme of underlying representations of the sentence, dominant in contemporary linguistic theory. Some meaning representation at the deepest and most abstract level undergoes various transformations until it reaches the surface level of the sentence as uttered or written and heard or read. An even closer correlate of that linguistic scheme would have the logical mechanisms a level lower than the script oppositions. The script opposition is equated with the meaning, or content, of the joke in this representation, and it is shaped at the lower levels by acquiring a logical mechanism, a target, and the language implementation.

This linguistics-inspired account presents a number of serious theoretical problems. First, yielding to Hofstadter's criticism (see Hofstadter and Gabora 1989 for the context), based on pure logic, the original proposal was modified by adding the logical mechanisms to the same level of abstraction as the script oppositions. An inelegant solution to start with, it also ran against the linguistic intuition which had informed the whole scheme in the first place. Second, by thus prying the scheme loose from linguistic theory, it opened a Pandora's box of problems concerning the ordering of the levels.

The reason the original proposal was revised to accommodate the script oppositions and logical mechanisms at the same level, rather than by adding a separate level for the logical mechanisms below the script oppositions, was that the two components seemed to be independent from and freely combinable with each other, and so the primacy of the one over the other could not be established. A somewhat similar situation exists on level 3. Originally termed "cultural instantiation," it was meant

to accommodate the selection of the target and situation for the joke, as shown in Table 1. The relationship between these two selections was not established, but their coexistence on the same level again implied mutual independence.

Very similarly to hierarchical models of representation in linguistics, in the original joke representation model, the determination of one level by another was the ground for placing the former under the latter, and the transitivity relation held as well. In other words, the implicit claim was that level 5 determined level 4 directly and each of the lower levels transitively, or indirectly; level 4 determined level 3 directly and levels 1–2 transitively, etc. (see Figure 2, in which the thicker arrows denote direct determination and the thinner transitive, indirect).

Before exploring further the interdependency of the levels in Figure 2, it is necessary to address yet another serious problem with this representa-

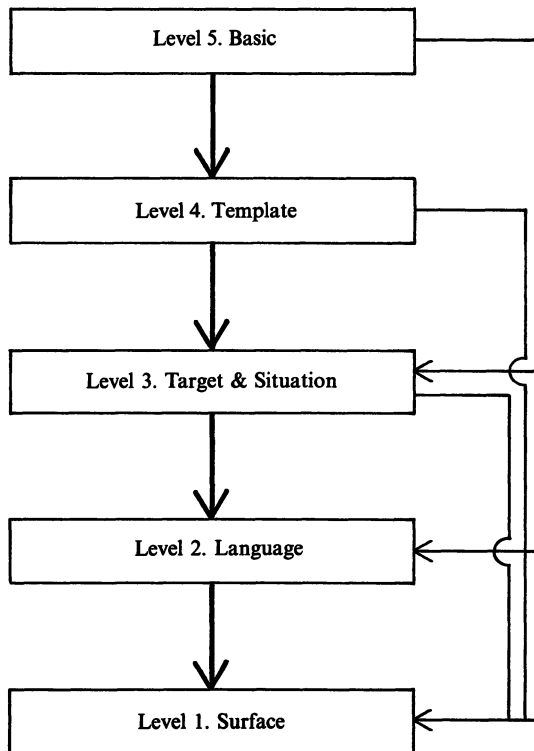


Figure 2. Attardo's five-level hierarchy

tion of the joke. The levels are not equal in their nature. Levels 5, 3, and 2 are substantive in that they contain bodies of knowledge upon which the joke draws, namely, the hierarchical list of script oppositions of various levels of generality as well as the list, possibly also hierarchical, of main and auxiliary logical mechanisms on level 5, massive cultural information of two different kinds, target-related and situation-related, on level 3, and massive linguistic information on level 2. Levels 3 and 2 are also operational in that a complex mechanism of selection fits the cultural and linguistic data, respectively, into the abstract representation of the joke. Level 4 is purely operational in that it simply combines the two bodies of knowledge of level 5. Level 1 is simply the output of level 2, and nothing else happens there.

This last problem could probably be rectified by the elimination of levels 4 and 1, assigning the operational function of the former to level 5, making the level 1 representation, that is, the actual final text of the joke, the output of the whole scheme, and supplying a "dummy" input to it for the sake of the input/output completeness, as shown in Figure 3.

This revision would render all the postulated levels similar in nature by making them all both substantive and operational. It would not, however, solve the problem of the uncomfortable coexistence of two distinct bodies of knowledge⁹ on each of levels 3 and 2 nor that of one missing body of knowledge informing the joke, namely the list of available narrative strategies.

Six knowledge resources

There may be a temptation to place the narrative strategies along with all the other language choices on level 1. After all, one may argue, this is just a form of expression. Level 1 already contains the choice of a natural language and of the whole set of pragmatic, semantic, syntactic, morphological, phonological, and phonetic language options to implement the joke, including — most importantly — the punchline. The choice of a genre is not, however, part of the same game, and, as shown in the preceding section again, it can be independent of the purely linguistic choices.

In other words, we have six bodies of knowledge, or knowledge resources (KR), or databases, which inform the joke, the script oppositions (SO), logical mechanisms (LM), situations (SI), target (TA), narra-

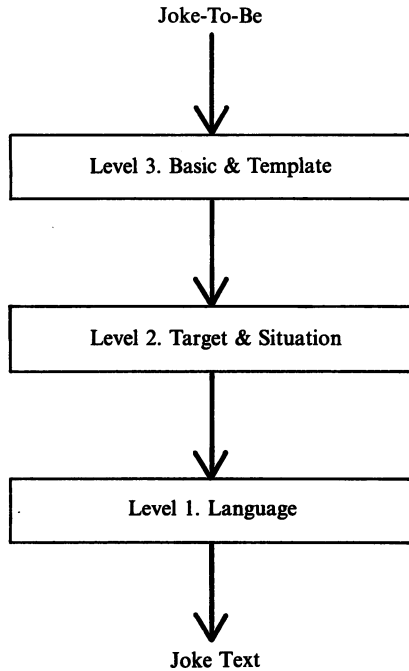


Figure 3. *Homogenized level hierarchy*

tive strategy (NS), and language (LA) (see Figure 4). Each KR is a list or set of lists from which choices need to be made for use in the joke. Now, we can address the issue of how these choices affect each other in hope that this will shed light on the order of abstraction, generality, and importance among the KRs.

The making of a theory: postulating a hierarchy

The six KRs in Figure 4 all enjoy an equal status, and their relations to each other are unknown. A hierarchical arrangement along the lines of Figures 2 and 3, on the other hand, would determine a strong ordering among the six KRs. Capturing all the existing bilateral and multilateral relations among the components is one advantage of a hierarchy as opposed to unordered equality. For a hierarchy to emerge, therefore, those relations should first be proven to exist, and we will proceed to discover and establish them in the next subsection.

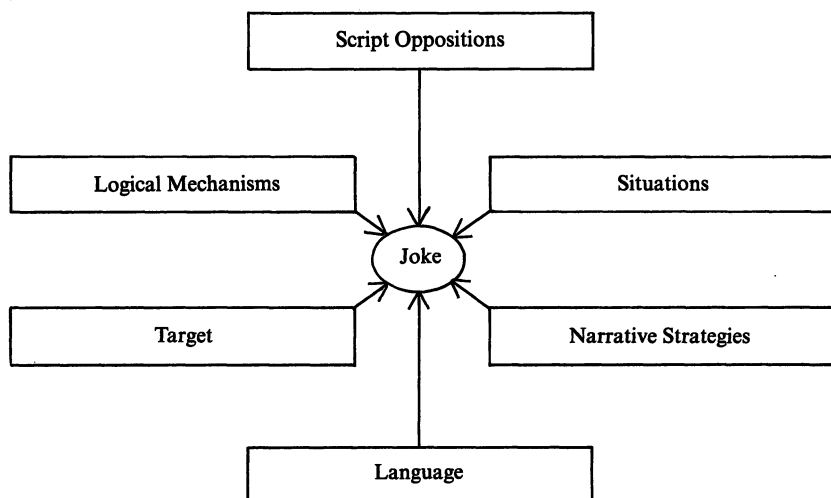


Figure 4. *Knowledge resources informing a joke*

If a hierarchy does result from a set of bilateral and multilateral relations among the components, it will provide an abstract model of joke generation in the strictly logical sense of the term *generation*, in which it was introduced into linguistics by Chomsky (1957, 1965). It means simply a convenient, dynamic representation of an entity, a joke in our case, as a process in which the decisions and choices about the various traits and ingredients are made in a justified logical order. What it does not mean is that jokes are actually produced this way by the speakers, as we will explore a little further in a later subsection.¹⁰

Even if not rooted in the actual process of joke production, a hierarchical representation of the components informing the joke would have gained enormously in validity if it did have some empirical roots, and another subsection below will argue that the degrees of similarity among the jokes, even if clear only in some cases and legitimately opaque in other, borderline cases, do provide this anchoring for the hierarchy which will emerge.

The issue of levels of abstraction has not yet been handled directly by the philosophy of science, even though research on theory formation in general has sharply increased in the last decade or so (see Curd 1980 for an informative survey of the logic of discovery as well as Thagard 1988: 65–70 on theory formation in general). The similarity-based anchoring, along with whatever little can be inferred on the subject from theory-

formation research, suggests two general principles for placing one KR above or below another.

Principle 1. The Roseanne Barr rule, or the wider you are, the higher up you go. If KR-X and KR-Y relate in such a way that a choice made within the former limits the choices within the latter, then X precedes Y in the hierarchy. Formally, it can be represented as follows:

KR-X choices:	{a,b,c ...}
KR-Y choices:	a1,a2,a3,...; b1,b2,b3,...; c1,c2,c3,...; ...}
Interlevel dependencies:	a→a1,a2,a3,...
	b→b1,b2,b3,...
	c→c1,c2,c3,...
	...

While *a1*, *a2*, *a3*, and other *a*-related choices may, in general, overlap with some *b*- or *c*-related choices, the pattern remains quite clear, namely, the choice of *a* on X reduces the number of possible choices on Y. Figure 5 introduces the funnel, the operational metaphor for principle 1, whose name is derived from the shape: the higher levels have a wider array of choices, the lower the narrower.

Gradually restricting the choices top-down, the abstract process of joke generation results in a single joke dropping through the narrow funnel opening.

The gradual top-down narrowing corresponds well intuitively to the relation of similarity: if, for instance, all the choices have already been made within KR1 through KR5, then the range of choices left on KR6 is very limited; some two jokes sharing all the KR1 through KR5 choices and differing only in the lower-level KR-related choice are

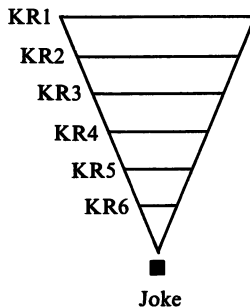


Figure 5. *Free-flow funnel: the choice-limiting hierarchical configuration*

likely to be more similar than some other pair of jokes differing in a higher-level choice, say in KR1. We will see below if this is really the case.

Principle 2. The Donald Trump rule, or stop the flow? — Down you go! If KR-Y rigidly determines KR-X, then KR-Y should follow and not precede KR-X in the theoretical model. By rigid determination, we mean that any choice made within KR-Y uniquely determines the choice made within KR-X. In other words, a choice within KR-X is precluded from being made at that level. This is what is shown in Figure 6a. KR4 restricts the KR5 and KR6 choices to one, thus rigidly determining them. What it actually means is that the KR5 and KR6 choices are subsumed by the KR4 choices, and, thus, KR4 simply absorbs KR5–6, and the three KRs would be conflated into one, as shown in Figure 6b. Given the different nature of each KR, such an arrangement would considerably impoverish and encumber the model. As it will be shown in the next section, this is exactly what SSTH does by skipping all but two KRs, SO and LA, thus absorbing the remaining four in the latter.

To avoid the constriction of the funnel, a KR like KR4 should follow rather than precede KR5 and KR6.

Ideally, an asymmetrical relation between KR-X and KR-Y, such that KR-X only reduces the choices on KR-Y but KR-Y rigidly determines KR-X, imposes a clear ordering of the two KRs, namely, with KR-X preceding KR-Y. The following subsection analyzes the binary relations among the six KRs.

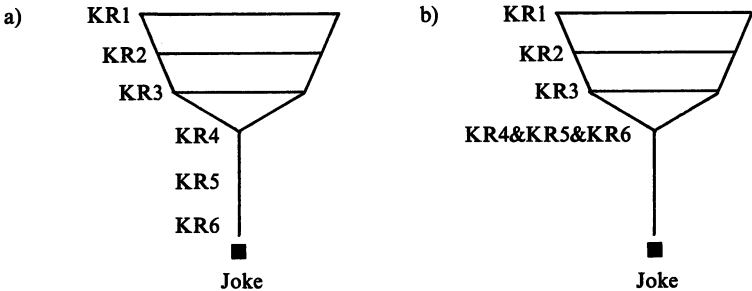


Figure 6. *Constructed funnel: one KR rigidly determining two others*

Binary relations among the KRs

To start with the easier relations, the text of the joke, which is the result of all the LA choices, embodies, implements, and reveals all the choices made with regard to the other five KRs. In other words, if the punchline is already in place and doing the job, nothing much is variable with respect to the other KRs. It is only natural to suggest, therefore, that the text of the joke emerges after all the other choices have been made. To assume the contrary, that is that the LA choices are made before any other KR choices, will be tantamount to eliminating the possibility of choice within that other KR. What seems to follow is that the other five KRs should precede LA (Figure 7). The order among those five KRs remains, however, to be determined.

It was mentioned earlier that SO and LM are independent of each other. Jokes (1) and (6) demonstrate that the same script opposition may be combined with different logical mechanisms. Jokes (21)–(23) show that the same logical mechanisms are compatible with different script oppositions; in fact, even the script opposition types are different in the three jokes, good vs. bad, real vs. unreal, and normal vs. abnormal, respectively. It is true, however, that the logical mechanism in (1) fits the script opposition better than the logical mechanism in (6), which is proven, among other things, by the fact that (1) is a real-life joke¹¹ while (6) has been artificially concocted for the purpose of easy comparison here. Such preferences should also be expected within other pairings of script oppositions and logical mechanisms. What matters, however, is that the nonpreferred combinations can work, too, so the two KRs retain

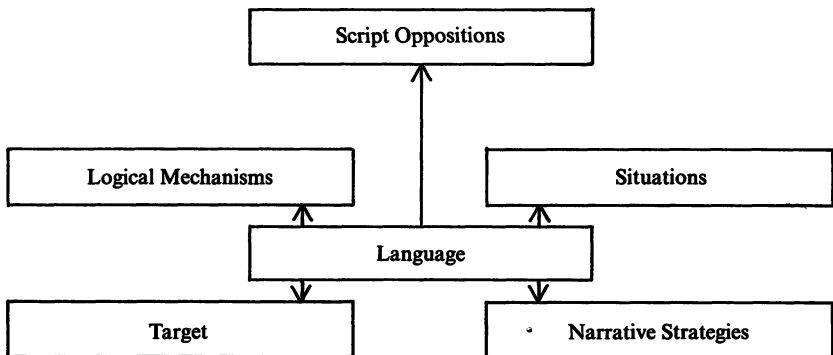


Figure 7. *The five knowledge resources determined by language*

their basic mutual independence. The preferences are analogous in a sense to stylistic preferences in a text.

Jokes (1), (5), and (14)–(17) demonstrate that the choice of a script opposition does not predetermine the situation, which varies from light-bulb changing to car washing to teeth brushing to fanning oneself to emptying an ashtray to driving, respectively. It can be argued perhaps that this particular script opposition, smart vs. dumb, puts a constraint on an acceptable situation: it should be one in which one can make a fool of oneself. It can be convincingly counterargued that one can make a fool of oneself in just about any situation. It seems to be equally plausible that a particular situation may be compatible with various script oppositions. The only two examples supporting that have been (1) and (7), but it is easy enough to concoct jokes based on other good/bad oppositions (such as nonstingy/stingy, noncunning/cunning, life/death, etc.) as well as on other types. SO and SI should then also be treated as mutually independent.

The opposite seems to hold between SO and TA. In order to be used with the smart/dumb script opposition, the target group (or individual) should be believed to be dumb. In other words, the dumb stereotype should be strongly and widely associated with the group.¹² Joke (1) would be incapacitated by replacing the Poles with the Jews because the latter have a set of totally different stereotypes associated with them (cunning, entrepreneurial, neurotic, etc. — see Raskin 1985: 209–221; Ziv 1986, 1991). The relationship between SO and NS as well as between SO and LA does not seem to exceed the level of stylistic preferences. Jokes (1) and (2), on the one hand, and (1) and (3), on the other, seem to bear this out.

An LM choice may constrain the SI choices, and vice versa, because, for instance, a situation may not have a clear figure and ground and thus be incompatible with the figure–ground reversal as a logical mechanism. LM does not, however, influence TA (compare jokes [1] and [4]) and may have only stylistic preferences with regard to NS (compare jokes [1] and [3]) or LA (compare jokes [1] and [2]).

Independent of SO and constraining LM, SI also constrains TA because the cultural baggage imported by the choice of target rules out the situations incompatible with it. Thus, the dumb stereotype excludes smart and sophisticated situations. Since brain surgeons are stereotypically associated in American culture with the highest degree of smartness, one should not expect to succeed with a joke involving a real board-certified

Polish brain surgeon. Jokes (1)–(3), again, show that only stylistic preferences can be implied by SI with regard to both NS and LA.

TA rigidly determines SO unless a certain targeted group can be associated with more than one stereotype within the same culture, generally a somewhat confusing situation which is preferable to avoid in joke telling.¹³ TA also constrains SI and LA, the latter only to the extent that the name and/or characteristics of the target should be reflected in the LA choices; for instance, if it is a Polish joke the word *Pole* or *Polish* should be present in the text (alternatively, though, it could be an obviously Polish name, as in joke [13], or some other unmistakably and popularly recognized Polish piece of *realia*). TA may imply only a stylistic preference with regard to NS (compare jokes [1] and [3]). It is timely to recall here that TA is an optional parameter.

NS may constrain some LA choices but will not at all influence SO, LM, SI, or TA. LA, on the other hand, will rigidly determine SO, LM, SI, TA, and NS, as argued above. The relations among the six KRs are shown in Table 2, with “I” indicating that the KR in the column is independent from the KR in the line, “S” indicating that the KR in the line may have a stylistic preference for the KR in the column, “C” indicating that the KR in the line constrains the KR in the column, and “D” indicating that the KR in the line rigidly determines the KR in the column.

The symmetrical relations among KRs in Table 2 cannot help us to establish any sort of ordering among them, but the asymmetrical can. Clearing Table 2 of the symmetrical values results in Table 3.

These relations of order among the six KRs follow then from Table 3, with “>” standing for “higher in abstraction/generalality” or “further from the text of the joke,” and “*” after KR standing for “any KR”:

KR* > LA

KR* (but LA) > NS

SO > TA

The resulting partial ordering of the KRs is shown in Figure 8. It is still quite removed from the linguistics-inspired completely ordered schemes in Figures 2 and 4.

Note that the arrows in Figure 8 have a different meaning from those in Figure 7. In the latter, the arrows indicate rigid determination as in Figure 6. In the former and subsequent charts for the emerging representation model, the arrows indicate the relations between the higher, more

Table 2. Relations within each pair of KRs

	SO	LM	SI	TA	NS	LA
SO		S	I	C	S	S
LM	S		C	I	S	S
SI	I	C		C	S	S
TA	D	I	C		S	C
NS	I	I	I	I		C
LA	D	D	D	D	D	

Table 3. Asymmetrical relations within pairs of KRs

	SO	LM	SI	TA	NS	LA
SO				C	S	S
LM					S	S
SI					S	S
TA	D				S	C
NS	I	I	I	I		C
LA	D	D	D	D	D	

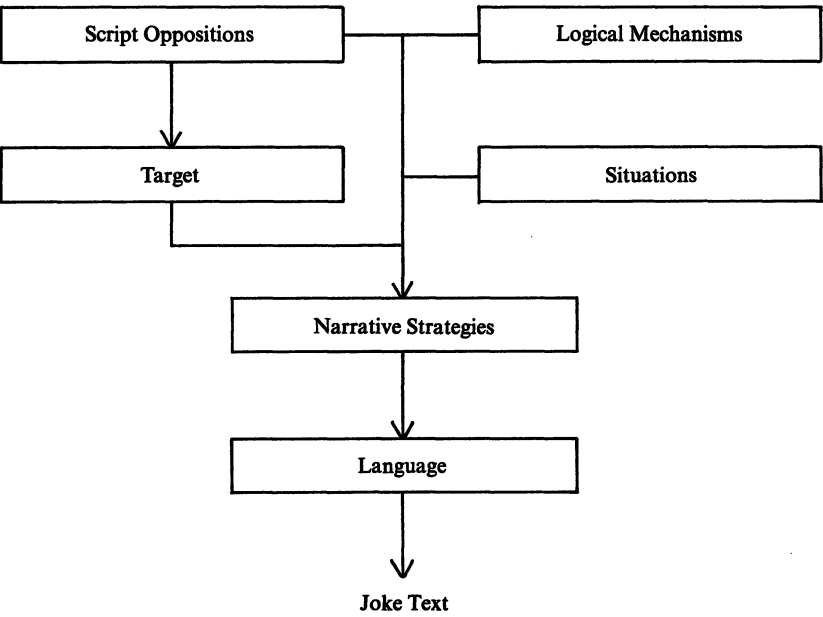


Figure 8. Partial ordering of the six KRs on the basis of the binary relations

abstract, preceding levels of representation and the less abstract, lower, following ones.

Content KRs and tool KRs?

It is possible to try yet another angle in order to achieve a complete linear ordering of the KRs, similar to the one in Figures 2–3. Can we perhaps distinguish between content-oriented KRs and tool-oriented KRs? Can one argue that joke (1) is about dumbness, Poles, and light-bulb changing rather than about figure–ground reversal, riddles, and language options? If so, then SO, TA, and SI are content-oriented KRs while LM, NS, and LA are the lists of tools which are used to express the content.

Two of the tool-oriented KRs have already been placed at the two bottom levels of abstraction, the closest to the text. Does it follow, then, that the third tool-oriented KR, LM, should be placed directly above NS while the three content-oriented KRs take the three upper rungs? If so, the still incomplete ordering shown in Figure 9 will result.

Joke similarity as basis for ordering?

In terms of the choices made within each KR, jokes (1)–(7) can be represented as shown in Table 4. LA 1 simply indexes the set of language choices made for joke (1). LA 2 denotes a deliberately different set of choices. LA 1* denotes a set of choices as close to LA 1 as possible under the circumstances; in other words, LA 1* contains the same options wherever possible with the minimal changes necessitated by one different choice within another KR. This notation further underscores the point made earlier in the section about LA being influenced by each other KR and rigidly constraining each of them in its turn. Ignoring the necessary (as opposed to optional, deliberately introduced) differences between LA 1 and LA 1* in each case of the latter's occurrence, we can say that each of jokes (2)–(7) differs from joke (1) in one KR choice only. All the other pairs of jokes (2)–(7) differ in two choices.

The strongest psychological evidence for a complete and linear ordering of the KRs in the theoretical model would come from a simple linear ordering of the degrees of similarity among jokes (1)–(7). First, it is necessary to establish that the degrees of similarity among jokes (2)–(7)

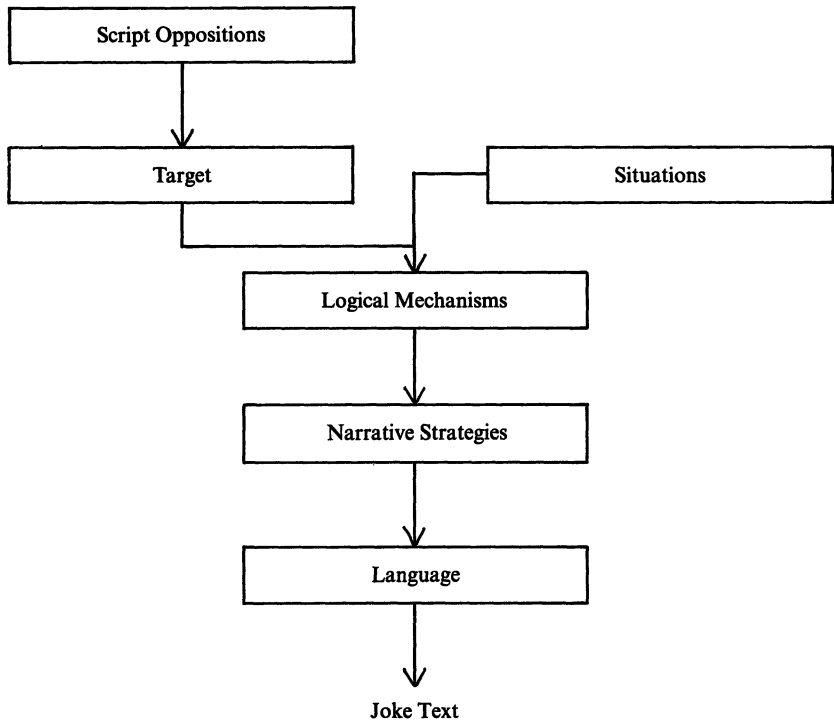


Figure 9. *Partial ordering of the six KRs on the basis of the binary relations and the content/tool dichotomy*

Table 4. *The KR values in jokes (1)–(7)*

Joke/KR	LA	NS	TA	SI	LM	SO
1	LA1	riddle	Poles	light bulb	figure-ground	dumbness
2	LA2	riddle	Poles	light bulb	figure-ground	dumbness
3	LA1*	expository	Poles	light bulb	figure-ground	dumbness
4	LA1*	riddle	Irish	light bulb	figure-ground	dumbness
5	LA1*	riddle	Poles	car wash	figure-ground	dumbness
6	LA1*	riddle	Poles	light bulb	false analogy	dumbness
7	LA1*	riddle	Poles	light bulb	figure-ground	dirtyiness

are all lower than the degrees of similarity of each of jokes (2)–(7) to joke (1). This would confirm that the number of different choices is reversely proportionate to similarity and thus somewhat indirectly confirm the validity of the KRs. Then it would be necessary to establish, for instance,

that *the order of jokes (2)–(7) in the text* of this article or, to put it differently, the relative physical proximities of the text of each of these six jokes to the text of joke (1), *corresponds to the degree of similarity of that joke to joke (1)*.

Such a result would be interpretable only on the basis of a strong but plausible assumption, namely, that the less difference is caused by a different choice within a KR, the less deeply, or lower, the KR resides within the theoretical model. In other words, the more significant changes are introduced (and the less similarity perceived) by the deeper, or higher, more abstract, and general KRs.

Thus, if joke (2) is more similar to joke (1) than any other joke among (3)–(7), then LA is indeed the least abstract and lowest KR, the closest to the text of the joke, as it was suggested in the previous subsections. If joke (7) is the least similar to joke (1), then SO is the highest, most abstract, deepest KR. Then, according to the italicized supposition above, all the other KRs would be arranged correspondingly (see Figure 10 below, but not quite yet).

But can these perceptions of similarity be validated? It may sometimes be hard to decide, in general, whether one similarity or one difference is greater or less than another similarity or difference. It is easier to accept the first of the two conjectures above, namely, that the two-difference pairs are less similar than the one-difference pairs, in other words, that any two of jokes (2)–(7) are less similar to each other than any one of them to joke (1).

As noted briefly at the end of section 1, jokes (1)–(3) are much more similar to each other than jokes (5)–(7) are to any one of them. Joke (4) is a borderline case: on the one hand, it differs from joke (1) in just one word; on the other hand, it is not a Polish joke. It is also hard to decide which of jokes (5) and (6) is more similar to (1). Our strong inclination is to treat (5) as much more similar to (1) because it is exactly the same joke involving a totally analogous situation. Joke (6) is a different joke about the same situation because it has a different resolution (and punch-line — the screwdriver instead of table-turning). Joke (7), however, seems to be the least similar of all.

It is not at all clear whether these intuitions can be corroborated in rigorously designed psychological experiments of the type conducted, for instance, by Peter Derks (Derks and Arora 1990). Chomsky implied (1965: 19), and we strongly concur, that native speakers may not have any clear-cut intuitions about borderline cases. Of course, he was talking

about linguistic intuitions, and we are extrapolating this observation to include humor perception and similarity perception. But we may fully expect to obtain highly diverse results on the similarities between jokes (1) and (4), on the one hand, and jokes (1) and (5), on the other. Either one of these two similarities may be considered closer by any number of informants. Such evidence would, of course, be one reason to treat the involved KRs as equi-high on a scheme like the one in Figure 10. Other reasons may, however, overturn this decision.

The hierarchical joke representation model in Figure 10 is based on the weak psychological intuitions about the degrees of similarity among jokes (1)–(7), and it is compatible with all the previous suggestions and reasons for them, with the exception of the idea to put all the tool-oriented KRs below the content-oriented ones (see Figure 9). A little trick may correct this aberration as well and make some sense in the process. Why not suggest that a tool-oriented KR may serve another individual KR or a few but not all of them? This would constitute a modified content/tool dichotomy. In its light, we can treat LM as the tool for SO only, while NS and LA will remain the tools for all. This will make the scheme more compatible even with the five-level model in Figure 2, which, in fact, put SO and LM next to each other on the same level and dedicated the next higher level to their interaction.

Joke production as basis for ordering? No!

The last question to answer in connection with the ordering of the six KRs in the emerging joke representation model is whether the actual process of producing, that is, uttering, a joke can shed some light on the hierarchy. What actually happens when we decide to tell a joke? Where do we start? What comes first? What happens next? What happens last? The last question is the only one which has an easy answer. What happens last is that the text of the joke is uttered. What happens before that depends on a whole slew of factors. First, there are two kinds of jokes, and the circumstances of their production are quite different.

The canned jokes (see Fry 1963: 43; Raskin 1985: 27; Mulkay 1988: 57–61) are reproduced from memory as a whole in an appropriate situation. The appropriateness may be provided by an association pertaining to virtually any one KR. Thus, one may witness a dumb act (SO), hear another ground-figure reversal story (LM), find oneself in a car wash

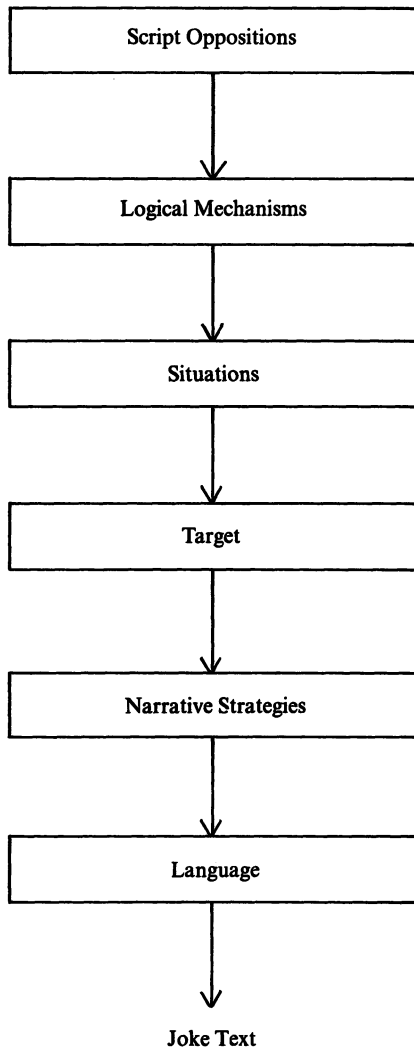


Figure 10. *A complete hierarchy of the six KRs on the basis of the binary relations, modified content/tool dichotomy, and similarity hypothesis*

(SI), hear a Polish joke (TA), participate in a riddle contest (NS), or hear a joke phrased exactly like that and be reminded of joke (1). One may also find oneself in a situation totally unrelated to any element of joke (1) and feel motivated to tell it, for instance when a group of people are

telling each other jokes and joke (1) is the one the speaker chooses or the only one he/she can think of at the moment.

The situation(al) jokes (see Fry 1963: 43; Raskin 1985: 27; Mulkey 1988: 62–66) are really produced out of their components. The situation provides some of the components explicitly or implicitly, and the producer of the joke provides the rest. According to SSTH (Raskin 1985: 140), the necessary components of the joke are

- i. a switch for the bona-fide mode of communication to the non-bona-fide mode of joke telling;
- ii. the text of an intended joke;
- iii. two (partially) overlapping scripts compatible with the text;
- iv. an oppositeness relation between the two scripts; and
- v. a trigger, obvious or implied, realizing the oppositeness relation.

Component (i) is, in fact, a precondition of the joke (see Raskin 1985: 100–104 and 141 for a detailed discussion). Component (ii) is trivially present in a verbal joke and incorporates component (v), the punchline. Components (iii) and (iv) pertain to SO. SSTH analyzes the various possibilities of observing some typically occurring combinations of (ii)–(v) in a situation and the techniques of providing the missing components. Thus, for instance, the most frequently occurring and “cheapest” combination is the cooccurrence of a potential trigger, especially of the most easily available one, the pun, and one script, which is the actually occurring situation.

As mentioned earlier, SSTH incorporates five KRs, LM, SI, TA, NS, and LA within SO and refraining from their further analysis. In other words, it goes straight from SO to the text implementing a particular script opposition. In view of the present discussion, the process of joke construction will, of course, involve all the KRs; it will, however, remain the same in principle, that is, to make a joke, one would need to observe some of its components in the current situation and to provide all the missing ones. Virtually any combination of KR factors may be present in a situation and cause the production of a joke by calling for the appropriate choices pertaining to the absent KRs. It seems intuitively more natural or perhaps more likely and, therefore, more frequent for a joke to be initiated by at least one content-oriented KR. However, it is perfectly possible to think of a situation in which a figure–ground reversal is observed, and this tool-oriented KR, perhaps in combination with some other factors, becomes the one which starts the joke off.

No matter what component starts the joke, the speaker is still responsi-

ble for all the others. And because virtually any one component may start the joke, it has no bearing on the hierarchy of the KRs in the theoretical model. If one starts with a lower KR, the joke can still be analyzed as generated top-down, with the starting KR inserted in the appropriate slot in the process. As studies in linguistic theory strongly suggested in the 1960s, generation as analysis and representation, on the one hand, and generation as actual production, on the other, may not, and possibly cannot, be served by the same model.

Thus, much of contemporary linguistics analyzes the syntactic structure of a sentence as the process of generation starting with the initial symbol S and passing many underlying levels of decreasing depth and abstraction and experiencing complex transformations, all before reaching the surface form of the sentence. It would be absurd to suggest that this is how a native speaker produces the same sentence because it would imply a desire to utter an abstract symbol S at the very beginning of the sentence. Instead the speaker starts off with some content to convey and perhaps some ready-made elements for the utterance that will eventually be made and then proceeds to supply all the missing elements without skipping a single phenomenon represented by the consecutive levels of representation.

This is the most important lesson of the analysis-vs.-production dichotomy: the subsequent levels of an abstract model of analysis do not correspond to the consecutive stages of actual production; that is, contrary to a naive expectation, the order of levels is totally devoid of any temporal value — a lower level is not a later level.

The joke-analysis-vs.-joke-production dichotomy seems to bear this out fully as well. In this sense, any reasonable joke-production model will be fully compatible with a reasonable hierarchy of KRs, such as shown in Figure 10. In fact, it is rather hard to visualize any production-related evidence that can bear in a meaningful way on the theoretical model because what occurs to the speaker first or what the speaker observes first is then washed out by all the other components which have to be provided.

A general theory of verbal humor

After having discovered the lack of any actual connection between the cognitive act of producing a joke and a reasonable ordering of its compo-

nents, this is a good place to reiterate the question asked earlier in the paper, why have levels at all? To emulate linguistic theory and thus satisfy the linguistically trained authors? So, what's wrong with that?! Nothing perhaps, if both the intrinsic value of the linguistics paradigm and its applicability to humor research are demonstrated or taken for granted. This is not, however, the real answer.

First, it is to be hoped that the different KRs represent accurately enough the various components of the joke. Distinguishing the components is analysis, and analysis is the basis of all theory. To the extent that KRs influence each other asymmetrically, an order can be established within each pair. A linear ordering of all the KRs is the simplest and neatest way to combine all the statements about the hierarchical relation within each pair. A model based on such a simple hierarchy is easier to use. However, the simplest and neatest is not always what is there, and a theory may have to do with a partial ordering, such as the one shown in Figure 8.

The discussion in the previous section provided enough justification, based on various dimensions, to accept, at least tentatively, the complete linear hierarchy of the six KRs as the levels of joke analysis shown in Figure 10. For the balance of the paper, we will assume the validity of that model, and each ensuing statement pertains to it as a proposal for a general theory of verbal humor (GTVH).

GTVH is fully falsifiable as any reasonable hypothesis/theory should be (see Popper 1972). It can be falsified on a purely theoretical ground by discrediting one of its KRs and replacing it with another as well as by simply adding a new KR to it. Such a falsification would amount to a theoretical revision of GTVH. It can be falsified on empirical grounds if humor research/psychological experiments discover a reality incompatible with the similarity-related conjectures of the preceding section, namely, if the subjects reliably and uniformly consider two-difference jokes more similar than one-difference jokes and/or if they consider pairs of jokes differing in higher KRs more similar than those differing in lower KRs.

Similarity needs to be addressed here again. Along with the multiple levels of joke representation, Attardo's five-level model introduced the idea of joke variants and invariants. The joke invariant was basically the combination of a particular script opposition with a particular logical mechanism. GTVH is compatible with this idea and, in fact, makes SO and LM the highest levels in the model. Attardo's level 4 joke template

was the juxtaposition of SO and LM {SO, LM}. This template is replaced in GTVH by a six-argument template for a joke, {SO, LM, SI, TA, NS, LA}.

Thus, GTVH offers a much more general, powerful, and versatile concept of joke variance and invariance. In GTVH, each variance relation is indexed by the argument KRs that the two jokes share. Thus, jokes (1) and (2) are {SO, LM, SI, TA, NS} variants; jokes (1) and (6) are {SO, SI, TA, NS, LA} variants; jokes (2) and (7) are {LM, SI, TA, NS} variants (see Table 4); jokes (1) and (25) are {SO, TA} variants.

The list of arguments the variants share defines the corresponding invariant. Thus, jokes (1) and (2) are both *dumb, figure-ground reversal, light-bulb changing, Polish riddles*, with the italicized text describing their five-argument invariant; jokes (1) and (6) are both *dumb, light-bulb changing, Polish riddles*; jokes (2) and (7) are both *figure-ground reversal, light-bulb changing, Polish riddles*; jokes (1) and (25) are both *dumb Polish jokes*. Similarly, (31) is a {SI} serial template, and all the jokes compatible with it, including jokes (1)–(4), (6)–(7), (9), (11), (13), and (19), are *light-bulb changing jokes*.

(31) How many _____ does it take to change a light bulb? _____ [number].

One to _____ and [number minus 1] to _____ (see Raskin 1985: 186).

Other serial jokes are just as easily accommodated with templates with one or more arguments fixed. An invariant is then one of the available values of one or more (up to five — a six-argument invariant uniquely defines a particular joke) arguments, or one of the available choices on one or more (up to five again) of the six KRs.

SSTH dealt with {SO} invariants. Attardo's five-level model privileged {SO, LM} invariants. Both approaches are perfectly justifiable within an appropriate context. In fact, it is very likely that the higher-level invariants like these two will turn out to be much more useful for generalizations in humor research than some accidental assortment of lower-level arguments.

GTVH incorporates, subsumes, and revises both SSTH and the five-level model. Both revisions are reasonably friendly. There is, however, one very significant difference in status between the GTVH and SSTH. The latter is a linguistic (semantic) theory of verbal humor (as represented by the joke — see Attardo and Chabanne forthcoming for a discussion

of the role of the joke as the preferred text type in humor research). It is an application of an independently motivated and developed semantic theory to humor research. GTVH is a much more general, less linguistics-based theory of verbal humor (again, as represented by the joke). It is an application of a linguistic theory and of many other contributions from a whole slate of other disciplines to humor research. Table 5 attempts to capture these contributions.

GTVH is a general and essentialist theory of verbal humor in the sense that it addresses the "what" question, that is, "what is humor?" It does not address a number of other possible questions, such as "why does humor exist?" or "how do people use humor?" Those questions are handled by special theories of humor. The "what" question is the same question as the one addressed by SSTH. It is also the question addressed only by the incongruity-based theories of humor rather than by either of the two other major groups of humor theories, the disparagement-based theories and the release-based theories. The first of these are special

Table 5. *The six KRs and the contributing academic disciplines*

KR	"Standard disciplines"	Specific areas of research
SO	linguistics philosophy anthropology psychology	semantics, pragmatics, discourse analysis metaphysics, ethics language and culture cognitive, social psychology, psycholinguistics
LM	linguistics psychology philosophy mathematics rhetoric	discourse analysis cognitive psychology logic, modal logic, fuzzy logic logic tropes
SI	philosophy sociology psychology	theory of action socioeconomic differentiation cognitive and social psychology
TA	anthropology philosophy sociology psychology political science history	ethnic groups, stereotypes ethics, stereotypes socioeconomic differentiation social psychology circulation of power history of ethnic relations
NS	literary studies folklore rhetoric	narrative studies, genre folkloric narration, fairy tale structure discursive practices
LA	linguistics mathematics computer science	all areas catastrophe theory [for the punchline only] artificial intelligence, computational linguistics

theories attempting to deal with the alleged communicative goals of humor; the other are special theories trying to penetrate the issue of the pleasurability of humor. Still other theories of humor may deal with the role of humor in society in general and in politics, business, or education in particular.

The fact that SSTH addresses the same question as the incongruity theories is often enough for SSTH to be automatically included within this class of theories. While not at all harmful for the theory, this automatic inclusion is substantively incorrect because many incongruity-based theories carry a conceptual baggage that SSTH has no use for, for instance, the psychological arousal-resolution theories. It is not clear either whether every single SO within SSTH can be easily classified as an incongruity—it is clear, however, that the simple negation and contradictions definitely can.

GTVH inherits the same reticent affinity to the incongruity theories. It is not very revealing, however, to consider it a member of the class not only for the reason mentioned above about SSTH but also because GTVH (as was SSTH) is much better defined, developed, and explicated than a regular incongruity-based theory and so cannot really benefit much from the purported resemblance.

Purdue University

Notes

- * [From the Editor:] This is the first time that an article authored by the Editor is being published here. In a journal with perhaps the tightest peer-review procedure in the industry, it has been essential to ensure the same kind of objective appraisal for this article as for any other submission. The alternative would be to avoid publishing the Editor's work in the journal, and neither the publisher nor, in fact, the Editor take kindly to the idea of the Board members' relevant work being published elsewhere because this *is* the premiere journal of humor research.

The regular peer review for a *HUMOR* article consists of

- the initial screening of the submission by the Editor for its appropriateness as a scholarly work in general and for the journal in particular;
- the appointment by the Editor of the monitor for the article from among the Board members;
- the appointment by the monitor of two independent readers who report their opinions to him or her;

- the monitor's report to the Editor, containing the readers' opinions and the monitor's synthesis and recommendation;
- the Editor's decision ranging from an outright rejection to acceptance "as is," with five intermediate stages involving revisions and the after-revision status;
- the author's revisions, if necessary; the Editor's decision as to the necessity of another round of peer review;
- the additional round, if necessary;
- the Editor's final decision.

The anonymity of the monitor and the two readers is, of course, assured even as their comments are delivered to the author.

In case of the Board members' submissions, a Board member other than the author functions as the monitor, anonymously for the author, and the procedure is not different from any other; at no point does the author participate in the publication decision. In the case of the Editor, however, the procedure has to be modified to minimize his participation. The fact of his submitting the article can be taken in lieu of his initial screening. At the very end, he can rely on the readers' consensus as to the publication decision. The most difficult part to avoid is the selection of the monitor.

In this case, I decided simply not to appoint one. Instead, I sent a draft to the three toughest readers known to me both from the 3.5 years of their monitoring and reading for the journal and from my familiarity with their editorial practices prior to and outside of the journal. All are Board members and prominent humor researchers who have never been taken in by the script-based semantic theory of humor. Each was offered anonymity, and an elaborate procedure was established for assuring it. None of them opted for it, and they all signed their comments. All voted to accept the article as is, with optional suggestions. We have accommodated a very significant part of these suggestions in the revision and replied to the rest in the notes.

It was precisely the fact that the readers chose to sign their comments which made it possible to add a few notes to the article debating some of the points in their letters. Besides presumably satisfying the critics better than leaving some suggestions, even if offered as highly optional, unanswered, this arrangement has enabled me to publish the colleagues' very interesting thoughts without waiting for these thoughts to be integrated into their authors' own published work. Besides, quite a few of these points were made outside of the referees' own field of interest, so they might have never appeared in print on their own. Also, our word in the debate does not need to be the last. The critics may choose to respond. I have always favored having a debate or letters-to-the-editor section in the Newsletter, and so have the two consecutive Newsletter editors. But the readership has been so deferential... Well, these readers will not be! So this may be a good way to start a forceful and expeditious exchange of views in print.

This arrangement for published criticism rebuttal is something which, I think, we should generally encourage. Quite a few authors write me careful and thoughtful rebuttals of the anonymous readers' opinions — many of

them use only printable words. Some of these rebuttals as well as the initial criticisms that cause them definitely deserve publication, and more often than not, they are not included in the revised drafts. When I see this happening in the future I will encourage the authors to weave some of this stuff into their end notes. I would be happy to hear from the readership about this suggestion. And we'll see how it works out for this article.

1. This is the first of our criticism rebuttals mentioned in the previous note. Before proceeding to the substantive matter, we would like to take this opportunity to thank the journal's internal readers for the article, Professors Mahadev L. Apte, John S. Morreall, and Elliott Oring, for their greatly appreciated input, most of which has been incorporated into this revised draft. We are also grateful to the volunteer readers, Professors Giovannantonio Forabosco, a Psychologist based in Ravenna, Italy, and Willibald Ruch of the University of Düsseldorf, for their suggestions. Needless to add, none of the above is responsible for any faults still remaining in this revised draft.

Now for the substantive issue at hand. The phrase "as represented by verbal jokes" has been added in response to a reader's remark. Morreall (1990) writes,

There's more verbal humor than humor based on semantics. You call the theory a general theory of verbal humor, but it's really a theory of jokes. It doesn't seem to cover such verbal humor as:

- funny rhymes like the weak internal rhymes of Ogden Nash
- funny excessive alliteration
- spoonerisms and other verbal slips which don't result in new unintended words
- funny abuses of morphemic patterns ("if it's feasible, let's fease it" or P. G. Wodehouse's "He may not have been actually disgruntled, but he was certainly far from grunted.")
- much humor based on pragmatic incongruity, such as the person whose answer to a question is woefully short or long, the child exaggerating or lying, etc.

I don't see that the SSTH in this paper can handle even all jokes, if you count as jokes such things as:

- Irish bulls like "Policeman: Say you! If you're going to smoke here, you'll have to either put out your pipe, or go somewhere else."
- funny sayings like:
 - ☐ "Everything tastes more or less like chicken."
 - ☐ "You can get anywhere in 10 minutes if you go fast enough."
 - ☐ "Eat a live toad first thing in the morning and nothing worse will happen to you the rest of the day."
 - ☐ "Do you realize that man is the only animal that chews the ice in its drink?" (Martin Mull).

If you have a definition of "joke" which excludes these, then they are still kinds of verbal humor not covered by the theory.

Morreall's lists are a very representative and impressive assembly of cases to which the application of the script-based semantic theory of humor is not as straightforward as in the case of simple jokes. Some similar examples are analyzed in Raskin (1985: 132–139). The theory can definitely handle everything on his second list, most simply the Irish bull. The last two jokes are parodies of popular, even if unspecified, gems of scientific or pop-psychological observation, and the previous two are of a similar nature. Parodies are allusive, and all that means in terms of scripts is that the specific scripts alluded to by the speaker be available to the hearer(s), at least as per the speaker's assumption.

The humorous units in Morreall's first list are of a metalinguistic nature. In such jokes, the actual language of the jokes is contrasted to the common, unrhymed, unalliterated, unspoonered, well-formed, and appropriate language (in addition, Nash's funny rhymes are, again, parodies of unspecified regular rhymes). This contrast is easy to deal with by extending the script theory into handling metalanguage by adding a dimension to the standard *normal vs. abnormal* opposition. Of all of these examples, alliteration strikes us as the most difficult to handle — but it is still manageable.

Thus, the modifier restricting verbal humor, which we have added to the sentence in the text to accommodate Morreall's comment, is not really necessary. The theory we are proposing and the script theory of which it is a revision can handle verbal humor besides what Morreall refers to as "semantic" jokes (by which he must mean something like "jokes easily handled by the script theory") and we label "simple" (meaning the same). By adding the constraint, we are simply playing it safe, highlighting the area of the highest performance for the proposed theory.

2. The two nouns in this last nominal phrase are both rather elusive. Without going into the "meaning of meaning" here, it can be argued that no two meanings are totally equivalent and that even a very slight change in the wording of a sentence always introduces a different shade of meaning. It is more realistic to talk here about a range of equivalence, that is meanings considered equivalent if they are sufficiently close to each other and no further away from each other than a certain threshold value. In other words, a clean and simplistic algebraic equation is inapplicable here; instead, we are talking about a fuzzy concept à la Zadeh (1965) and McCawley (1981) and a prototype à la Rosch (1973) and Lakoff (1987).

The prototype theory (see, for instance, Lakoff 1982, 1987) claims that some members of the set are "better examples" of that set. Dealing with the set of all birds Rosch showed that speakers tend to characterize robins and sparrows as the "best examples" of birds and rate penguins as the worst (see Lakoff 1982). The individual differences between speakers as to what they consider a bird, the color red, or a verb render these and, arguably, all categories fuzzy. The important point is the cutoff, at which an element is no longer a member of the set, an animal is not a bird, a color is not a shade of red, a word is not a verb.

The extension of the concept of paraphrase to the joke suggests such a

cutoff point of the joke sameness. Jokes (1) and (2) are seen as two members of the set of all paraphrases of this joke, that is, all the different ways the joke can be told without effecting any other changes in its contents. Some paraphrases may be closer to each other, others more remote, but all of them remain within the same range. It is therefore, strongly implied that jokes (1) and (2) are much more similar to each other than any other pair among jokes (1)–(7). This will be reiterated and discussed in detail in the next section.

3. As it will be shown in the next section, the other parameters provide input for this one by fixing certain elements in the contents of the joke as a given and obligating the language parameter to express it. Thus, if a joke is based on alliteration, the fixed elements that this parameter will need to express will indeed include the “lowly” phonetics. In such a case, any phonetic change affecting the intended alliterative effect may change the punchline and the joke, thus taking the resulting version far beyond the range of the paraphrases of the joke.
4. Two of the three principal readers were troubled by this term and one of them by the substance behind it as well. Thus, Oring (1990) writes,

Is the strategy you are discussing always a “narrative” strategy? While I know what you mean I wonder if “narrative” is really the term you want unless you are using narrative in some special sense. It seems jarring to label a riddle or question [and] answer structure a “narrative.”

Both terms, “strategy” and “structure,” he uses are appropriate here. A narrative strategy assigns a certain macrostructure to the text of the joke. It assigns the joke to a certain mode of presentation. The preceding parameter of language takes it from there. The narrative strategy is one of the things that cannot be changed under the parameter of language.

Apte is concerned about the division of labor between these two parameters. He writes (1990),

I am not convinced that the knowledge resources of language and narrative strategy are independent of each other, not at least in verbal humor. Why is choice of a genre not a part of the same game?... It seems to me that throughout this paper the term “language” is used in two senses: 1) ...as a system of communication; and 2) [as] style. In the latter sense it is not distinguishable from the... “narrative strategies.” For instance,... [the difference in the language between jokes (1) and (2)] seems to me to be more a contrast of “style” and thus of the “narrative strategy” used in the generation of the joke.

The misunderstanding is natural not only because of the tentative and indeed special nature of the term “narrative strategy” but also because of the tendency on the part of the linguists, of whom we are two, to use the terms in carefully defined technical senses and the tendency on the part of nonlin-

guist readers to understand these terms in much wider, extended, and unintended meanings. The term "language" in the preceding parameter is not meant either as a system of communication or as style. It is meant to include the mechanism of language activated for the expression of certain contents. With the exception of very small-scale decisions, such as the choice between two synonyms or the difference in meaning between two paraphrastic syntactic constructions (for instance, between a conjoined sentence and a sentence with a relative clause or between the latter and two separate sentences), all the crucial content-related decisions are made under the auspices of the other parameters. The genre — and style — of the text of the joke are two of these decisions, and they are made under the auspices of the parameter of narrative strategy. It should be noted that Apte and the other readers saw an earlier draft, in which the concepts of language and narrative strategy were much less explicitly described. It is because of Oring's and Apte's difficulties with narrative strategy that the extensive revision and elaboration of these two subsections have been undertaken.

5. Apte again: "Why is non-redundancy only a part of this parameter and not also of the parameter 'language'?" This is easier to explain in view of the previous note. The nonredundancy, leading to implicitness, is an important element of the presentation of the contents of the joke and, therefore, a significant part of the narrative strategy. The opposite decision would be the deliberate excessive verbosity of the presentation (see Morreall's "woefully...long" answer in note 1); this would be a secondary decision under this parameter, more compatible with some major decisions, such as the choice of the expository narrative strategy, than with others — can there be a verbose riddle?
6. Always on the alert, Apte (1990) wants to know what these practical descriptive reasons are. Fair enough, even though it will become clearer and easier to explain in the next section. It is possible that the phenomenon of implicitness comes under the auspices of both this parameter and the parameter of logical mechanism. Because it will be shown that the latter precedes the former in the joke representation model deriving from the theory proposed in this article, it is much more theoretically prudent to assign implicitness to a lower parameter, which can take input from a higher one, than to assign it to a higher one and thus to preclude input from a lower one. The proposed solution makes the cooperation between the two parameters on this concept possible; the opposite decision would force an unnecessary choice of just one of the two parameters as the sole master of implicitness.
7. Concerning our use of "logical mechanisms," Forabosco (1990a) notes that both the term "logical" and the phenomena covered by it may be rather unclear. Our usage of the term "logic" in "logical mechanism" was motivated by the fact that whatever little discussion of these issues is available in the scientific literature (see references in the text) often uses terminology and/or notation from mathematical logic or their metaphorical extensions. Technically speaking, these mechanisms are probably "cognitive," but this term is no less confusion-prone than "logical."

On the issue of the content, Forabosco first discusses a very general kind of figure/ground reversal:

In the article, the critical issue seems to be the types and levels of the logical mechanisms considered. I'll concentrate on the figure-ground reversal mechanisms. ...*Per se* it does not belong to logic, but is configured as a perceptive phenomenon (Gestaltic), identified as that particular procedure that happens in the organization of visual stimuli (cf. Rubin's cup, etc.). ... Bateson, Fry, and others have used this phenomenon to describe "analogically" what happens in the organization of information (in communicative analysis). ... In this sense, the figure-ground reversal describes a very general phenomenon that obtains each time one or several elements, previously not considered, are put into direct attention.

Forabosco distinguishes another type of figure-ground reversal at a different level of abstraction: "There is also a restricted, denotative way of seeing the figure-ground issue in humor, and that is the exploitation of the peculiarities of the perceptive phenomenon as such."

The third level of figure-ground reversal, according to Forabosco, is even less abstract than the first — it is the "relational inversion" and it covers relations such as "turning the hand" vs. "static ceiling" in joke (1). This is the sense intended for the term in this article.

Forabosco is right about the potential ambiguity of the term as well as when he suggests that, in the first sense of the term, this specific logical mechanism would be common to almost all jokes, since all of them would involve a foregrounding of a previously dormant, "not previously considered" script. It is important to emphasize that we do not intend the "figure/ground reversal" logical mechanism broadly but rather as a mnemonically convenient label for a REAL VS. UNREAL role swapping for some elements in the joke situation. In this sense, what we utilize is a reasonable metaphoric extension of the intuitively clear spatial meaning of the term used by Talmy (see the references in the main text) in cognitive linguistic research. It should also be noted that with the other logical mechanisms identified in this subsection (such as negation, chiasmus, false priming), this kind of confusion seems to be less likely, if at all possible.

8. The term "justification" in this meaning was introduced by Aubouin (1948) as a technical term denoting one of the two basic aspects of a humorous fact. The two aspects are "incongruity" and "acceptance/justification." Aubouin notes correctly that two incongruous objects *per se* are not necessarily perceived as funny. In order to be perceived as humorous, the two objects have to be "accepted" simultaneously. Acceptance describes the behavior of the hearer, justification the "condition in which the object can elicit this acceptance" (Aubouin 1948: 95).

The nature of this process of acceptance, or justification, is defined by Aubouin as very brief, "superficial," masking for an instant the absurdity of the judgment (1948: 95). This acceptance does not enjoy the same logical

status as the acceptance of a mathematical proof (1948: 94). But how can a subject accept two incongruous ideas, even if briefly, and superficially? According to Aubouin, only by incurring in an "error of judgment," or at least, by accepting the possibility of an error and, thus, in our terms, imposing a faulty logic, or pseudologic, or paralogic (see also Forabosco 1990b) on the situation.

Aubouin divides these errors into two classes, the errors of judgment and those connected with language. He lists a large number of possible causes of errors of judgment, including similarities, bad conditions for observation, lack of experience, prejudice, fatigue, hasty generalizations, etc. The errors "of language" are similarities of sounds (which include homonyms and paronyms, that is, words with similar but not identical surface representations), ambiguities (semantic and syntactic), and literal interpretation of frozen metaphors, puns, alliterations, and a few other techniques.

In spite of this quite insightful discussion, Aubouin stops short of recognizing all these elements as the constituents of special faulty logic, legitimate for humor. Instead, he seems to believe that it is necessary for the hearer to be really misled by the humorous text. In other words, Aubouin links humor to a factual error on the hearer's part. This is not at all the case. There are cases in which the hearer will be truly misled at first, such as in the garden-path example (24), but in most cases, the hearer will be perfectly aware that he/she is violating an ordinary logic rule in a "willing suspension of disbelief." The example of puns is a case in point.

Consider joke (29). The hearer is aware that *strike* was first misleading used in the meaning of "catching one's attention" and later in the meaning of "hit." The two meanings of *strike* belong to sufficiently disparate semantic fields to be clearly identifiable as incongruous. How do these two disparate ideas get brought together, even if momentarily? This is possible only if the hearer accepts the fact that the two meanings of *strike* are superficially and brittly lumped together in a temporal sequence: one of them occurs first, and the other replaces it later. Clearly, only a conditional and very shortlived acceptance is possible because the hearer is well aware that the two meanings are distinct and mutually incompatible.

Because this skill which is brought into the understanding of puns is based on the speaker/hearer knowledge of the language, it has been claimed that puns have a "metalinguistic" aspect (Hausmann 1974: 8–10). It is interesting to note that one of the aspects of metalanguage is that the rules of the object-language are "suspendable" in the metalanguage. Thus, the sentence *The sentence COLORLESS GREEN IDEAS SLEEP FURIOUSLY is considered grammatical by Chomsky* is perfectly acceptable, even though it contains an ungrammatical (capitalized) part. With this in mind, Aubouin's claim that an actual error needs to occur is not only false but also unnecessary: if puns involve a metalinguistic reflection, the rules of languages are suspended legitimately and errorlessly, and, as a result, for instance, the two meanings of *strike* are temporarily reconciled.

9. "Why is coexistence of two distinct bodies of knowledge at the same level

'uncomfortable'?" asks Apte. And more generally, "Is hierarchy the most significant and necessary condition for any analysis?" Both questions pertain to the issue of theory formation which is dealt with later in the section. Unfortunately, the philosophy of science focuses primarily on natural sciences, and even there, it largely shies away from discussing the postulation of levels of abstraction and their ordering.

The two questions are indeed closely related. The first one is a little easier to handle. If a level of representation is postulated at all, why go to all the trouble only to end up with an unwieldy two-bodied monster on one's hands? An abstract construct should be neat and manageable, and a heterogeneous level like levels 2 or 3 is definitely neither. The only reason to put two very different entities on the same level was that they could not be shown to be hierarchically arranged: that is, neither is "higher" or "lower." This may be reason enough to give up the idea of a hierarchy but certainly not enough to declare the two entities part of the same whole.

This brings up the issue of a hierarchy. A case for it is made in small installments throughout this section. Here, we will provide a general foundation for these installments. Several entities may belong to a set, for instance, {a, b, c, d, e, f}, and in this case they are unordered and all of the same status, and not much can be known about their binary and other relations. If there is no interest in these relations on the part of a theoretician, the set may be an adequate model of representation. It is certainly not vacuous because it brings with itself the logical baggage of set theory, complete with its axioms and theorems. Even more importantly for formal approaches, it makes a rigorous system of discourse available for use.

If, however, the relations among the entities are of an interest to the theoretician, a more powerful concept of vector or cortege is evoked. In a vector, all the elements are linearly ordered, for instance, $\langle a, b, c, d, e, f \rangle$, with a preceding all the others, b preceding all but a , and so on. In terms of the precede-follow relation, a vector establishes a complete and simple order over the set — for each pair of elements, the vector determines explicitly which precedes which. The vector is especially good as a model for a one-feature ordering of the entities.

More complex and messier models of partial ordering are provided by postulating subsets within the set or combining vectors and sets within the same model. It makes sense to attempt a simpler linear model before resigning oneself to a more complex partial one. This is precisely the nature of the enterprise here. The single feature on which the proposed hierarchy is going to be based, the feature which is represented as the precede-follow relation among the levels of the model, is the feature of one level influencing another in terms of restricting (but not constricting) the choices made on the latter. This is what the balance of this section is all about. See also the next note.

10. Morreall writes,

The five-level joke representation model tries to get respectability from its similarity to transformational-generative grammars of languages. But

unlike TG grammars, this theory doesn't provide algorithms which generate jokes. It's more like the stratificational grammars which were supplanted by TG grammars. This theory makes no claim to psychological reality, certainly not for joke-telling, and not even for joke-creation. You deny that the levels and the "components informing the joke" are temporally ordered.

In those negative respects, of course, the theory is like TG grammars. But TG grammars can at least compensate for their lack of psychological realism by claiming predictive and explanatory power: a successful TG grammar will generate all and only those sentences of a language. This theory of jokes does not have well-defined algorithms which generate all jokes and only jokes. It's an abstract model without much predictive or explanatory power — much, as I said, like the stratificational grammars which TG grammars supplanted. Will the theory explain anything beyond the fact that certain pairs of jokes seem more similar than other pairs?

We are in a somewhat delicate situation here. We disagree with Morreall on a few counts in his assessment, and we can show that the theory we are developing here is much more like TG than he thinks it is. On the other hand, we are not keen on strengthening the link between our theory and TG because we want the former to stand on its own and not because this is the way things are done in linguistics and the authors happen to be linguists.

In fact, the only problem Raskin had with Apte's seminal position paper on humorology in the very first issue of this journal (1988) was that the author attempted to equate the issues and evolving methodology of humor research with those of his own field, anthropology. For him, that equation strengthened and validated his argument. For his nonanthropologist readers, it did not. On the contrary, the issues and the methodology argued for in Apte's article are much more persuasive when justified and motivated within the field of humor research itself and allowed to stand on their own, as they actually were, than when an attempt is made to validate the same issues and methodologies in another field, even an important and actively contributing one.

This is one pitfall we are trying hard to escape — we are not going to say, and we sincerely do not think, that something is good because it is good for linguistics. Such an argument would only alienate a nonlinguist. In fact, we have an immediate proof of that in Apte's and Oring's views of this article. Apte's question cited in the previous note reveals that he is not willing to take the hierarchy, this *sine qua non* of a linguistic model, for granted (a Ph.D. in linguistics notwithstanding!). His fellow anthropologist and folklorist Oring understates benevolently, "I am not always as excited by some of the formal gymnastics in this paper as a linguist might be (gymnastics you are at some pains to explicitly defend)."

In spite of these scruples, we still feel like defending our theory against Morreall's beliefs that it is not similar enough to the generative paradigm dominating linguistics in the last three decades. We actually believe that it

has all the good qualities of that paradigm and fewer bad ones. We also believe that two qualities that it shares with the generative paradigm and that Morreall, a philosopher with impressive credentials in linguistic theory, considers negative, are not.

Our theory claims no "psychological realism" in the sense of postulating the existence of the six KRs in the mind of the native speaker as they are construed in the theory. In a much weaker sense, however, it does enlist some (hypothesized) psychological evidence as one of the bases for its own justification; to put it more simply, it demonstrates what kind of psychological evidence (pertaining to the degrees of perceived similarity among jokes) would support the theory.

Our theory claims its neutrality to the process of "joke creation" in the sense that the precede-follow relation among the KRs is devoid of the temporal value; to put it more simply again, it is not claimed that higher-level decisions are actually made earlier by the joketeller. These two constraints are indeed shared by the theory with the "pure" generative paradigm in linguistics, free of Chomsky's and others' occasional "forgetfulness" about them (see Raskin 1976 for further discussion; there is also a devil's advocate argument there that TG as a production procedure might not be as indefensible as it might seem — if applied correctly, which it never was). These two constraints clear the theory from extraneous claims it cannot comfortably maintain and free it to be what a theory must be, namely, the general basis, format, and template for analysis.

This is precisely what both TG and our theory do. By analyzing each joke with the help of a reverse-analysis generative procedure, they provide a resolution mechanism (not really an algorithm in either case) for separating a well-formed sentence from a non-well-formed sentence in the case of TG and a (potential) joke from a nonjoke in the case of the proposed humor theory. It is this resolution function which renders both theories explanatorily and descriptively adequate and gives them the "predictive power," which means simply that they can predict what entities will and what will not belong to the privileged set (of well-formed sentences and jokes, respectively).

The generative paradigm and the humor theory also share the view that the phenomena they study are rule-governed. The former has been partially successful in presenting some of the rules in a relatively simple formalized notation. The latter has been partially successful in presenting many of the rules underlying jokes less formally though not less rigorously. We believe that it is this difference which misleads Morreall into thinking that TG has proposed an algorithm of the kind that the humor theory has not. In fact, both claims are incorrect: TG has not proposed — nor did it aspire to propose — an algorithm; whatever TG has proposed, the humor theory has proposed as well — and neither has practically implemented their respective proposals in full.

Less significant for this readership but still worth a passing mention is Morreall's belief that stratificational grammar is TG stripped of its explanatory and predictive power. It is true that, like all pregenerative linguistics,

stratificational grammar lacks those powers. It is also true that, again like all pregenerative linguistics, stratificational grammar postulates levels of linguistic structure. All of that stratificational grammar takes for granted and comes nowhere close to the concern about its own theoretical foundations inherent to the generative paradigm and borrowed (in its most significant part) by the proposed humor theory.

It should be reiterated again that these theoretical concerns with multibased self-justification, the status of postulated levels, the ordering among them, and so on are introduced in this article into humor theory for their own stand-alone appeal. They go beyond the generative paradigm in linguistics, and their partial genesis in that paradigm is almost accidental.

11. Apte makes an interesting point about this term. He writes,

I find the distinction made between a “real-life” joke and one “artificially concocted” dubious and unconvincing. Aren’t most verbal jokes artificially concocted? They may or may not later become “real-life” jokes, but it seems to me that we are really talking here about the quality or degree of opposition on which the joke is based and which has nothing to do with “real-life.” This is more a distinction between a “good” joke and a “bad” one, and that too, based strictly on the criterion of script opposition. Perhaps I misunderstand your use of the term “real-life.”

The point is well taken. On the other hand, being linguists again, we attribute significance to what actually occurs in speech as opposed to an artificially constructed example to illustrate a point. A real-life joke is indeed created by somebody mysterious at a certain point, but then it passes an important acceptability test by being memorized and repeated by people and thus becomes “real-life.” It certainly has something to do with quality, and the quality is indeed based on the kind of script opposition used, but then how can we explain the abundance of terrible, primitive, inane jokes based on squalid oppositions? Perhaps one explanation is that all the components of the joke should fit together better than a certain threshold value, which is probably not the case with joke (6).

12. Oring writes,

I don’t entirely agree. Jokes often establish the stereotype where none exists (in fact, I think that one of your students made such a point in her article on the information bearing nature of joke texts). Thus one could tell clever Pole jokes to someone who didn’t know any Polish jokes at all. Even if they knew the dumb jokes they might still recognize an out-of-stereotype joke. Although it would not be the best type of joke, it could still be recognized as a joke and would need to be addressed by your theory.

It is, of course, true that a hearer unfamiliar with the stereotype on which a joke is based but capable of recognizing the text as a joke can, in fact,

infer and thus acquire the stereotype. Yan Zhao (1988), the student referred to by Oring, addresses the issue from the early SSTH point of view. Her findings agree with Oring's assessment that, in their information-conveying function, jokes can still be recognized as such, but she goes further to claim, correctly, we believe, that in this capacity jokes are no longer funny.

13. Oring again:

But there are jokes about groups with a number of stereotypes. Jews are wealthy, canny, pushy, neurotic, wimpy, victims. All are used in jokes. Indeed, one could argue that such jokes would be better because there is more work involved in selecting the appropriate stereotype operating in the joke.

Generally, only one of the several available stereotypes about a target is activated in a joke. If two such stereotypes clash in a joke or if it is not immediately clear which of them is evoked, the joke will definitely suffer. More work is definitely involved in sophisticated jokes, and we share Oring's predilection for those. However, more than a certain threshold amount of work will kill the joke for many hearers, and still more will kill it for all. This is a very intriguing and almost totally unexplored subject. One route to pursue it is perhaps via a broad study of sophistication and its expression in language in general and in humor in particular (see Raskin 1990a, 1990b, and in preparation).

References

- Apte, Mahadev L.
 1988 Disciplinary boundaries in humorology: an anthropologist's ruminations. *Humor* 1(1), 5–25
 1990 Internal review of Attardo and Raskin 1991 for *Humor*. Unpublished manuscript.
- Attardo, Salvatore
 1987 Unpublished manuscript, Catholic University of Milan, Milan, Italy. (Published 1989 as Morfologia della barzelletta. *Studi Italiani di Linguistica Teorica ed Applicata* 3.)
 1988a Trends in European humor research: toward a text model. *Humor* 1(4), 349–369.
 1988b Type-jokes and token-jokes. Paper presented at WHIM-VII, the 7th National Conference on Humor. Purdue University, West Lafayette, IN, April 1–4. (Published 1990, in Hughes, Shaun F. D., and Victor Raskin (eds.), *WHIMSY VII: Proceedings of the 7th National Conference on Humor*. West Lafayette, IN: Purdue University.)
 1989 A multiple-level analysis of jokes. Contributed section in Hofstadter, Douglas, and Liane Gabora, Synopsis of the Workshop on Humor and Cognition, 438–439. *Humor* 2(4), 417–440.
 1990 The violation of Grice's maxims in jokes. In K. Hall et al. (eds.), *Proceedings*

- of the Sixteenth Annual Meeting of the Berkeley Linguistics Society. Berkeley, CA: University of California.
- Attardo, Salvatore, Donalee H. Attardo, Paul Baltes, and Marnie J. Petray
forthcoming The linear organization of jokes: an analysis of two thousand texts. *Humor* 5
- Attardo, Salvatore, and Jean-Charles Chabanne
forthcoming Jokes as a text type. In Attardo, S., and J.-C. Chabanne (eds.), special issue on humor research east of the Atlantic. *Humor* 4(4)
- Aubouin, Elie
1948 *Technique et Psychologie du Comique*. Marseille: OFEP.
- Bateson, Gregory
1953 The position of humor in human communication. In Von Foerster, H. (ed.), *Cybernetics*. New York. (Reprinted 1969 in Levine, J. [ed.], *Motivation in Humor*. New York: Atherton, 159–166.)
1955 A theory of play and fantasy. *A.P.A. Psychiatric Research Reports* 2. (Reprinted 1972 in his *Steps to an Ecology of Mind*. New York: Ballantine, 177–193.)
- Bloom, Harold
1965 Chiasmus. In Preminger, A., F. S. Warnke, and O. B. Hardison, Jr. (eds.), *Princeton Encyclopedia of Poetry and Poetics*. Princeton, NJ: Princeton University Press, 116.
- Chomsky, Noam
1957 *Syntactic Structures*. The Hague: Mouton.
1965 *Aspects of the Theory of Syntax*. Cambridge, MA: MIT Press.
- Clements, William M.
1969 The types of the Polack joke. *Folklore Forum. A Bibliographic and Special Series* 3.
- Curd, Martin V.
1980 The logic of discovery: an analysis of three approaches. In Nickles, T. (ed.), *Scientific Discovery, Logic, and Rationality*. Dordrecht and Boston: Reidel, 201–219.
- Davies, Christie
1990a An explanation of Jewish jokes about Jewish women. *Humor* 3(4), 363–378.
1990b *Ethnic Humor around the World: A Comparative Analysis*. Bloomington, IN: Indiana University Press.
- Derks, Peter, and Sunjay Arora
1990 Motivation, salience, and incongruity in humor appreciation. In Glazer, M. (ed.), *Abstracts. The Eighth International Conference on Humor*. Sheffield: University of Sheffield, 29.
- Esar, Evan
1952 *The Humor of Humor*. New York: Horizon.
- Fonagy, Ivan
1982 He is only joking (joke, metaphor and language development). In Kiefer, F. (ed.), *Hungarian Linguistics*. Amsterdam: Benjamins, 31–108.
- Forabosco, Giovannantonio
1990a Internal review of Attardo and Raskin 1991 for *Humor*. Unpublished manuscript.
1990b La logica dello humour. *Le Scienze* (Italian edition of *Scientific American*) 265(45), 60–66.

Freedman, Matt, and Paul Hofman

- 1980 *How Many Zen Buddhists Does It Take to Screw in a Lightbulb?* New York: St. Martin's.

Freud, Sigmund

- 1963 (1905) *Der Witz und seine Beziehung zum Unbewussten*. [Jokes and their relation to the unconscious]. New York and London: Norton.

Fry, William F., Jr.

- 1963 *Sweet Madness*. Palo Alto, CA: Pacific Books.

Glazer, Mark (ed.)

- 1990 *Abstracts. The Eighth International Conference on Humor*. Sheffield: University of Sheffield.

Hausmann, Franz J.

- 1974 *Studien zu einer Linguistik des Wortspiels. Das Wortspiel im Canard Enchaîné*. Tübingen: Niemeyer.

Hetzron, Robert

- 1991 On the structure of punchlines. *Humor* 4(1), 61–108.

Hofstadter, Douglas, and Liane Gabora

- 1989 Synopsis of the Workshop on Humor and Cognition. *Humor* 2(4), 417–440.

Hughes, Shaun F. D., and Victor Raskin (eds.)

- 1990 *WHIMSY VII: Proceedings of the 7th National Conference on Humor*. West Lafayette, IN: Purdue University.

Katz, Jerrold J., and Jerry A. Fodor

- 1963 The structure of a semantic theory. *Language* 39(1), 170–210. (Reprinted 1964 in Fodor, J. A., and J. J. Katz (eds.), *The Structure of Language: Readings in the Philosophy of Language*. Englewood Cliffs, NJ: Prentice-Hall, 479–518.)

Kerman, Judith B.

- 1980 The light-bulb jokes: Americans look at social action processes. *Journal of American Folklore* 93, 454–458.

Lakoff, George

- 1982 Categories: an essay in cognitive linguistics. In Linguistic Society of Korea (ed.), *Linguistics in the Morning Calm*. Seoul: Hanshin, 140–193.
- 1987 *Women, Fire, and Dangerous Things: What Categories Reveal about the Mind*. Chicago: Chicago University Press.

Marino, Matthew

- 1988 The good, the bad and the beautiful. *Humor* 1(1), 39–48.

McCawley, James D.

- 1981 *Everything That Linguists Have Always Wanted to Know About Logic*. Chicago and London: University of Chicago Press and Blackwell.

Mel'čuk, Igor A.

- 1974 *Opyt teorii lingvističeskix modelej "Smysl ↔ Tekst"* [An essay in the theory of the meaning ↔ text linguistic models]. Moscow: Nauka.

Miller, Arthur G. (ed.)

- 1982 *In the Eyes of the Beholder: Contemporary Issues in Stereotyping*. New York: Praeger.

Milner, George B.

- 1972 Homo ridens: toward a semiotic theory of humor and laughter. *Semiotica* 5, 1–30.

Morreall, John S.

- 1990 Internal review of Attardo and Raskin 1991 for *Humor*. Unpublished manuscript.

- Mulkay, Michael
1988 *On Humor: Its Nature and its Place in Modern Society*. Cambridge: Blackwell.
- Oring, Elliott
1984 *The Jokes of Sigmund Freud: A Study in Humor and Jewish Identity*. Philadelphia: University of Pennsylvania Press.
1989 Between jokes and tales: on the nature of punch lines. *Humor* 2(4), 349–364.
1990 Internal review of Attardo and Raskin 1991 for *Humor*. Unpublished manuscript.
- Pepicello, William A., and Thomas A. Green
1984 *The Language of Riddles: New Perspectives*. Columbus: Ohio State University Press.
- Popper, Karl R.
1972 *Objective Knowledge: An Evolutionary Approach*. Oxford: Clarendon.
- Raskin, Victor
1976 Generation and performance. *Linguistics* 181, 45–61.
1985 *Semantic Mechanisms of Humor*. Dordrecht: Reidel.
1986 Script-based semantics. In Ellis, D. G., and W. A. Donohue (eds.), *Contemporary issues in Language and Discourse Processes*. Hillsdale, NJ: Erlbaum, 23–61.
1990a Sophisticated jokes. In Hughes, Shaun F. D., and Victor Raskin (eds.), *WHIMSY VII: Proceedings of the 7th National Conference on Humor*. West Lafayette, IN: Purdue University.
1990b Sophistication in humor and beyond. In Glazer, Mark (ed.), *Abstracts: The Eighth International Conference on Humor*. Sheffield: University of Sheffield, 77.
in prep. *Semantic Mechanisms of Sophistication*.
- Rosch, Eleanor (Heider)
1973 Natural categories. *Cognitive Psychology* 4, 328–350.
- Samuels, Frederick
1973 *Group Images: Racial, Ethnic and Religious Stereotyping*. New Haven, CT: College and University Press.
- Schutz, Charles E.
1989 The sociability of ethnic jokes. *Humor* 2(2), 165–177.
- Spencer, Gary
1989 An analysis of JAP-baiting humor on the college campus. *Humor* 2(4), 329–348.
- Talmy, Leonard
1975 Semantics and syntax of motion. In Kimball, J. P. (ed.), *Syntax and Semantics*, vol. 4. New York: Academic Press, 100–120.
1983 How language structures space. In Pick, H., and L. Acredolo (eds.), *Spatial Orientation: Theory, Research, and Application*. New York: Plenum, 120–140.
- Thagard, Paul
1988 *Computational Philosophy of Science*, Cambridge, MA, and London: MIT Press.
- Todorov, Tzvetan
1978 Le mot d'esprit. In his *Les genres du discours*. Paris: Seuil, 283–293.
- Zadeh, Lofti A.
1965 Fuzzy sets. *Information and Control* 8, 338–353.

Zener, Walter P.

1970 Jokes and ethnic stereotyping. *Anthropological Quarterly* 43, 93–113.

Zhao, Yan

1988 The information-conveying aspect of jokes. *Humor* 1(3), 279–298.

Ziv, Avner

1984 *Personality and Sense of Humor*. New York: Springer.

Ziv, Avner (ed.)

1986 *Jewish Humor*. Tel Aviv: Papirus.

1991 Special issue on Jewish humor. *Humor* 4(2).

